



ABES Engineering College, Ghaziabad

Course Title: Basics of Mechanical and Automation Engineering

Course Code: To be decided

L T P

Course Credits: To be decided

2 1 0

Minimum Required Hours (Theory): 40 Hrs

Pre-requisites: Basic knowledge of Physics and Mathematics

Course Level: Undergraduate

Course Objective:

The course aims to build interdisciplinary awareness and practical insight into modern engineering applications. It provides all engineering students with a strong foundation in key concepts of Mechanical Engineering and Automation, applicable across domains such as design, manufacturing, thermal, energy systems, automation, and control engineering.

Course Outcomes: After the completion of this course, students will be able to

Course Outcomes		K Level
CO1	Apply the principles of engineering mechanics and analyze mechanical systems using concepts of force, stress-strain and material properties.	K3
CO2	Analyze thermodynamic systems and apply the laws of thermodynamics to basic energy systems, including HVAC applications.	K3
CO3	Explain the construction, working principles and comparative aspects of IC engines, electric vehicles, and hybrid vehicles.	K2
CO4	Describe the working principles and functions of sensors, actuators, and control elements used in basic automation and mechanical actuation systems.	K2
CO5	Identify and interpret the applications of robotics and 3D printing in smart and automated industries.	K2

Content	
Unit I	Introduction to Mechanics
Engineering Mechanics: Force moment and couple, principle of transmissibility, Resultant of force system- concurrent Types of supports (Hinge, Roller) and loads (Point, UDL, UVL), free body diagram, equilibrium equations and Support Reactions.	
Strength of Materials: Normal and shear Stress, Strain, Hookes' law, Poisson's ratio, elastic constants and their relationship, stress-strain diagram for ductile and brittle materials, factor of safety.	
Unit II	Basic Thermodynamics
Thermodynamics: Introduction to Thermodynamics, system and surrounding, basic thermodynamic terminologies (equilibrium, state, property, process, cycle), reversible and irreversible process, laws of thermodynamics (with numerical on first law of thermodynamics), comparison among heat engine and heat pump.	
Refrigeration: Refrigerating Effect, Ton of Refrigeration, Coefficient of Performance, Principle of Vapour Compression Refrigeration System, Construction, and Working of Domestic Refrigerator.	

Air-Conditioning: Introduction and Applications of Air Conditioning, Psychrometric Properties and Construction and Working of Window Air Conditioner.	
Unit III	Internal Combustion Engine & Electric Vehicle
IC Engine: Definition and Basic Components of an Engine, Construction and Working of Two-Stroke and Four-Stroke Spark and Compression Ignition Engines, Comparison Between Two-Stroke and Four-Stroke SI and CI Engines, Types of fuels. Electric vehicles and hybrid electric vehicles: Working principle and components of an EV, advantages and disadvantages of EVs. working of hybrid electric vehicles, series and parallel types of HEV, degree of hybridization, advantages and disadvantages of HEV.	
Unit IV	Introduction to Automation and Control
Sensors and Transducers: Introduction to Sensors, transducers and their applications, basics of Feedback and Control: Open-loop vs Closed-loop Systems. Overview of Mechanical Actuation System: Overview of kinematic chains, cam mechanisms, ratchet mechanisms, gears and their types, belts, and bearings. Hydraulic and Pneumatic Actuation Systems: Overview of fluid power systems, Pressure and direction control valves, rotary actuators, basic hydraulic and pneumatic sequencing problems.	
Unit V	Emerging Trends in Engineering
Robotics: Introduction and significance in automation, Classification (articulated, cartesian, cylindrical robots, etc.); Robot specifications (DOF, payload capacity, reach, repeatability), Introduction to SCARA robots. 3D Printing: Introduction and types of 3D printing technologies, materials used (plastics, metals, composites), 3D printing process workflow, applications and limitations.	

Text Reading:

- 1) Engineering Mechanics by S.S. Bhavikati & K.G. Rajasekharappa, New Age International (P) Limited
- 2) Strength of Materials by R.K. Rajput, S. Chand
- 3) Engineering Thermodynamics by R. K. Rajput, Laxmi Publications
- 4) Refrigeration and Air Conditioning by R.S. Khurmi, S. Chand
- 5) Internal Combustion Engines by V. Ganesan, McGraw-Hill Education
- 6) Electric and Hybrid Vehicles by Amir Khajepour, Wiley Publications.
- 7) Industrial Automation and Robotics by Mikell P. Groover, Pearson Education
- 8) 3D Printing and Additive Manufacturing by Ian Gibson, David Rosen, Brent Stucker, Springer

Reference Reading:

1. Engineering Mechanics by S. S. Bhavikatti, New Age International Publications.
2. Strength of Materials by S.S. Rattan, McGraw-Hill
3. Refrigeration and Air Conditioning by C.P. Arora, McGraw-Hill
4. Electric and Hybrid Vehicles by Iqbal Husain / M. Ehsani, CRC Press / Springer
5. Fundamentals of Electric Vehicles Technology by K.T. Chau, CRC Press
6. Mechatronics Systems by Rolf Isenmann, Springer.
7. Introduction to Industrial Robotics, by Nagrajan, Pearson India.
8. Robotics and Industrial Automation by R. K. Rajput, S. Chand Publication

Unit-1

Basics of Mechanical and Automation Engineering (BMA)

Contents : PART-1

#

Engineering Mechanics and Beams

- ① force and its types
- ② System of forces (types of force system)
- ③ Moment of a force
- ④ Force Couple
- ⑤ Principle of transmissibility
- ⑥ Resolved parts of a force
- ⑦ Free Body Diagram
- ⑧ Resultant of a concurrent (coplanar) force system
- ⑨ Analytical method to find resultant of several "Coplanar" concurrent forces.
- ⑩ Condition of static equilibrium in case of a coplanar concurrent force system.
- ⑪ Lami's theorem.
- ⑫ BEAMS — Types of loads on a beam
- ⑬ Types of supports
- ⑭ Types of beams
- ⑮ Problems based on calculation of support reactions.

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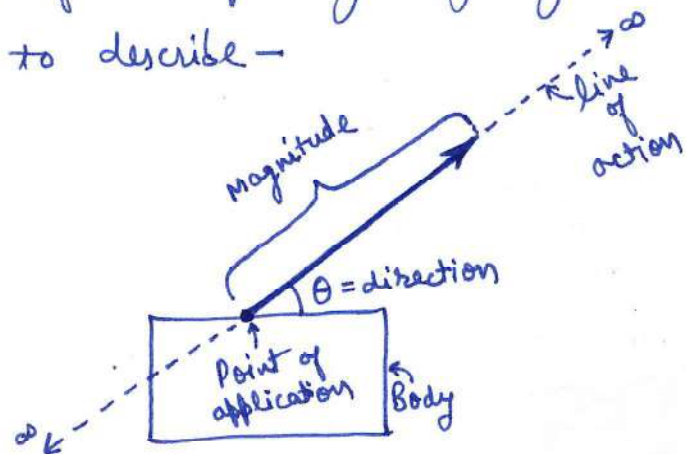
① Force and its types: Force is that external cause ~~which~~ due to which the magnitude or direction ~~of a body~~ of velocity of an object can change. It can create following changes in a system —

(a) can create linear displacement or angular displacement.

(b) can create linear/angular acceleration/deceleration.

Force is a vector quantity, and for completely defining a force, four things are necessary to describe —

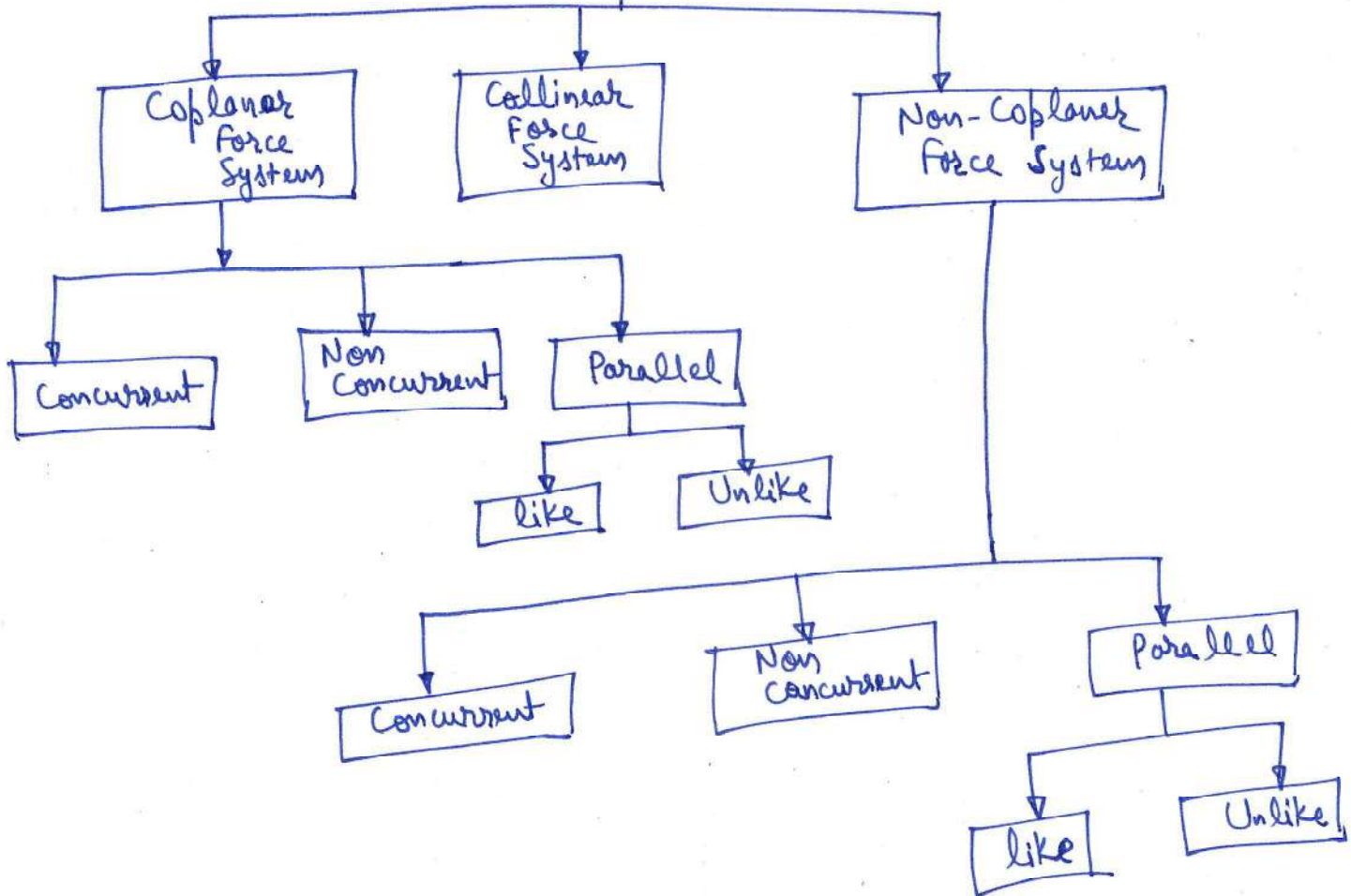
- (i) Its magnitude
- (ii) Its direction
- (iii) Its point of application
- (iv) Its line of action.



A force may be either 'push' type, or 'pull' type. Accordingly, it can also be considered as 'Tensile' or 'compressive' types.

② System of forces: It is a combination, or collective arrangement of more than one forces, acting in a system. The system of forces are classified according to following three diagram —

Types of force System

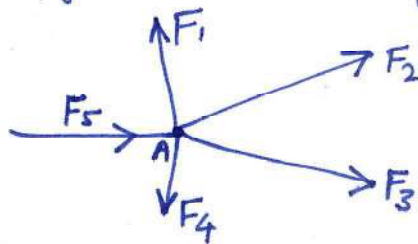


- Collinear Force System :- When the line of action of forces are in the same line (coinciding).

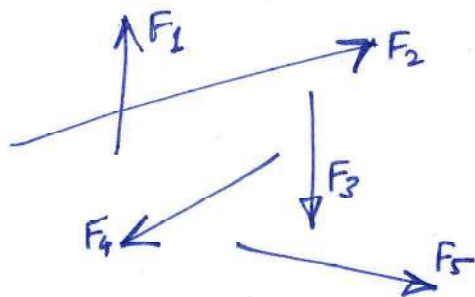


- Coplanar Force System :- When all forces are in a single plane.

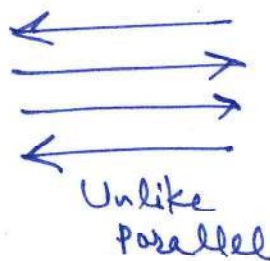
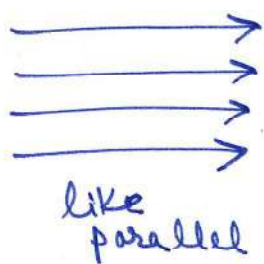
- Coplanar Concurrent force System :- When line of action of all coplanar forces are passing through a single point



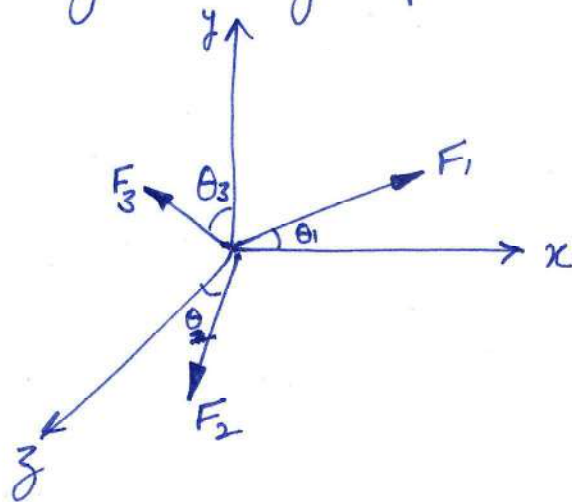
- Coplaner Non-concurrent force system :- When the line of action of all the coplaner forces do not pass through a single point.



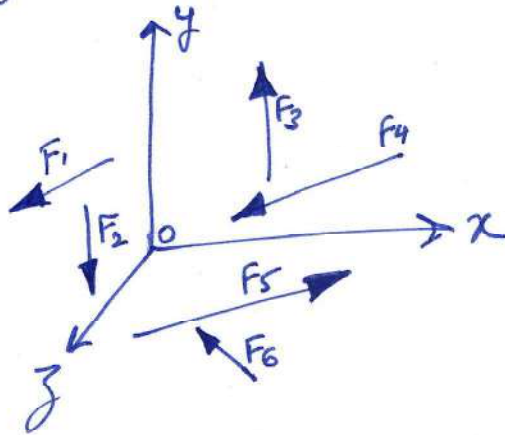
- Coplaner parallel force system :- When all the coplaner non-concurrent forces are parallel. There are two types -
 (a) like parallel
 (b) Unlike parallel



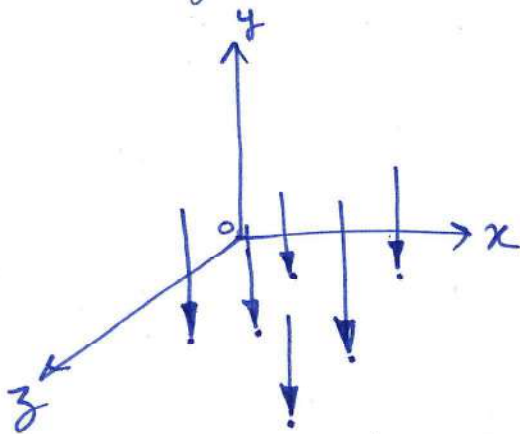
- Non-coplanar force system :- When the line of action of all the forces in a system are Not in a single plane.
- Non-coplaner concurrent force system :- When the line of action of all forces are not in a single plane, but all are passing through a single point.



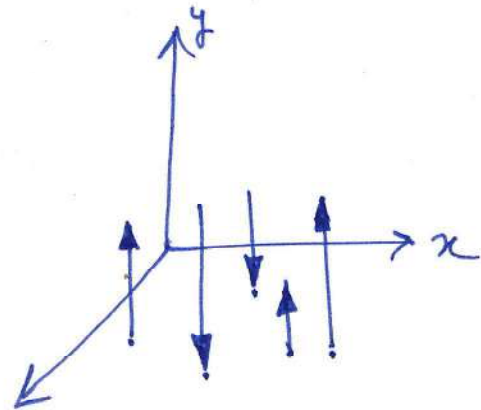
- Non-coplanar non-concurrent force system: When the line of action of all the forces are neither in a single plane, nor passing through a single point.



- Non-coplanar parallel force system: When the line of action of all the forces are not in a single plane, but all are parallel to each other. These are further of two types — (a) like, and (b) Unlike.



Non-coplanar,
like parallel.

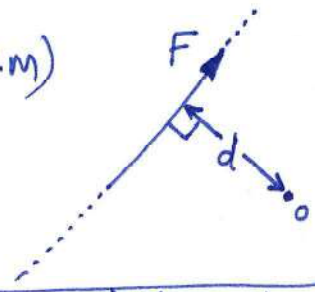


Non-coplanar,
Unlike parallel

- ③ Moment of a force: It is the rotational tendency created by a linear force, about a particular point. Moment may cause bending in Beams or structural members. It is measured by multiplying the magnitude of force with the distance of its line of action from a point.

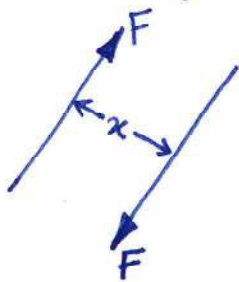
$$\text{Moment (M)} = F \cdot d \text{ (N.m)}$$

Note: It is different from the term 'torque' because torque is just like a rotational force, which can create twisting in a rod, and causes angular acceleration/deceleration.



④. Force Couple (or, only "Couple"):

When two equal and opposite forces act at a gap, then collectively they create a turning effect about their central point. This turning or rotating effect is called as "Couple", and is measured as multiplication of any one force and the distance between them.

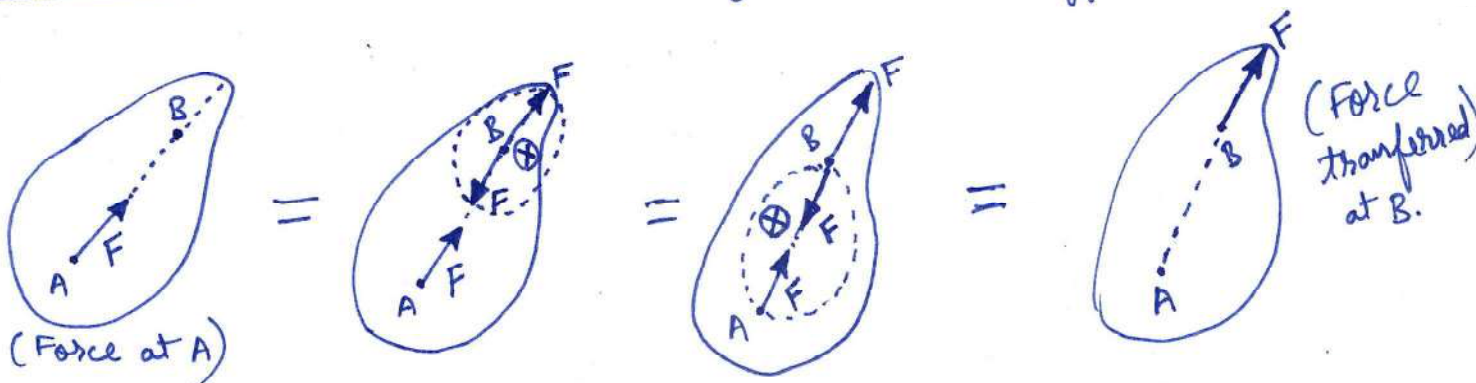


$$\Rightarrow \boxed{\text{Couple} = F \cdot x} \text{ (N.m)}$$

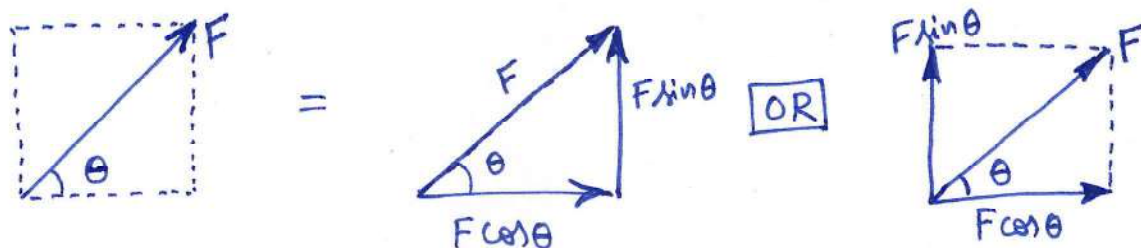
It may be either clockwise or anticlockwise.

⑤. Principle of Transmissibility:

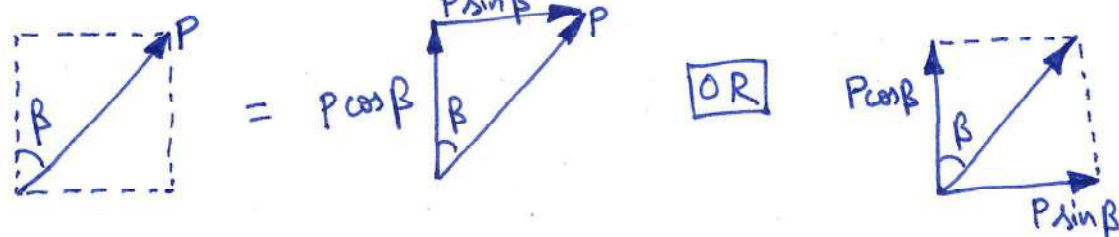
It is one of the most fundamental law of analyzing various force systems. It states that, "Any force can be transferred anywhere in its line of action within the limits of the object on which it is being applied. There will be no change in its effect."



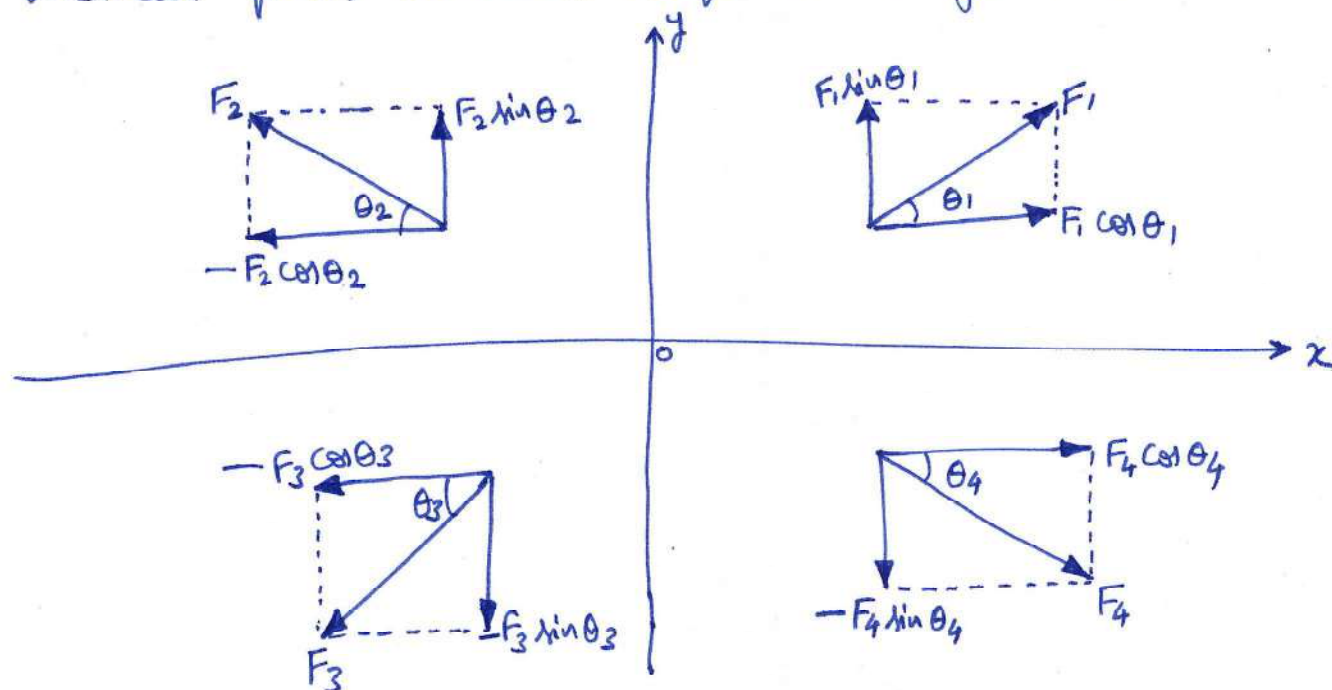
⑥ Resolved parts of a force : Whenever required, a force can be broken into two resolved parts, perpendicular to each other.



Similarly,



Any force can be resolved into a horizontal and a vertical part, whenever required as given below:



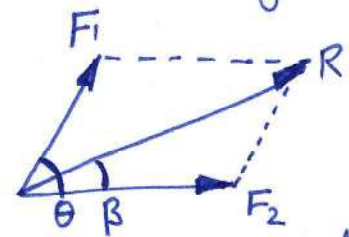
⑦ Free Body Diagram :- It is the diagrammatic representation of all the external forces, self-weight and support reactions acting on an object, or portion of an object. Here, all the supports, contact surfaces are removed and support reactions and contact reactions are shown instead.

Note: A Free Body Diagram can be drawn for an object under static equilibrium, or under dynamic conditions too.

⑧ Resultant of a concurrent (coplanar) force system

Two forces at an angle θ are acting at a point O. Then, their resultant can be calculated using the following formula-

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$$

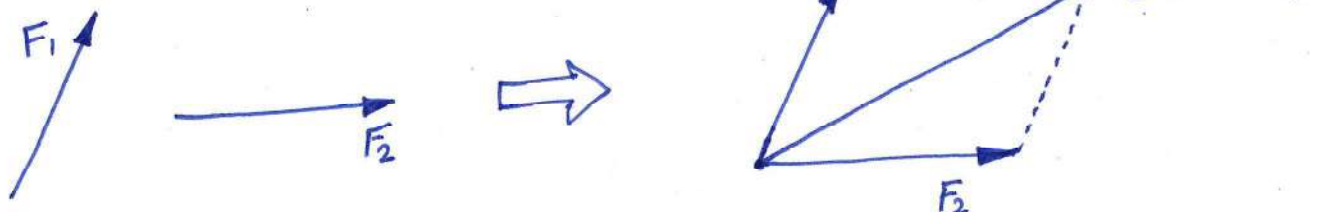


and the angle of the resultant with F_2 can be calculated-

$$\beta = \tan^{-1} \left(\frac{F_1 \sin \theta}{F_2 + F_1 \cos \theta} \right)$$

This method of finding the resultant is called analytical method. Another one is graphical method, or vector method: Parallelogram law of forces and Triangle law of forces (for concurrent two forces).

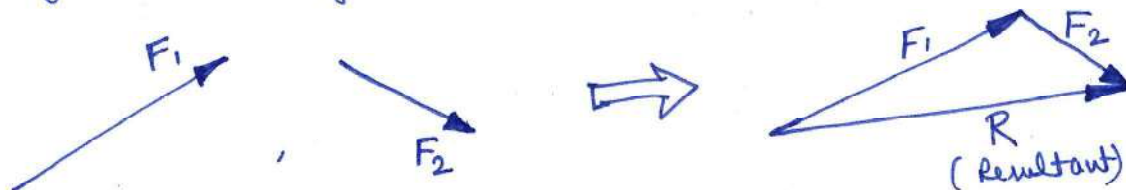
• Parallelogram Law of forces: —



"If two forces acting simultaneously on a particle, be represented in magnitude and direction by two adjacent sides of a parallelogram, then their resultant will be represented in magnitude and direction by diagonal of the parallelogram

which passes through their point of intersection."

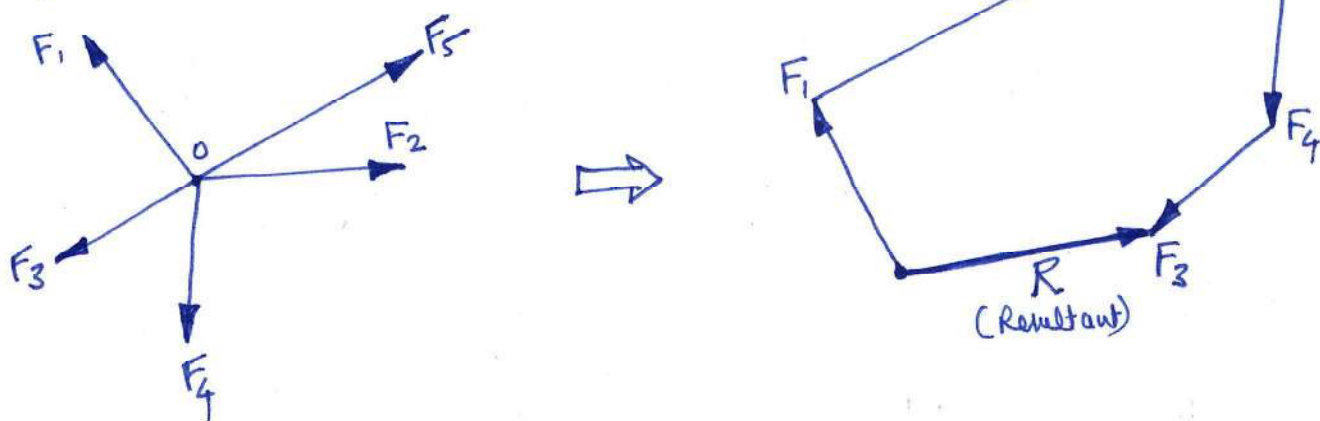
• Triangle law of forces:-



"If two forces acting simultaneously on a body are represented in magnitude and direction by two sides of a triangle taken in order then their resultant will be represented in magnitude and direction by the third (closing) side taken in opposite order."

Above two laws are applicable for two forces only. In case of more than two forces, polygon law of forces is used.

• Polygon Law of forces:-



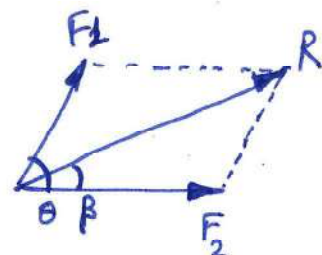
"If a number of coplanar concurrent forces, acting simultaneously on a body are represented in magnitude and direction by the sides of a polygon taken in order, then their resultant may be represented in magnitude and direction by the closing side of a polygon, taken in the opposite order."

⑨ Analytical method to find resultant of several "coplanar concurrent" forces :-

We already discussed earlier that the resultant of two forces at an angle θ is given by

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$$

$$\angle \beta = \tan^{-1} \left(\frac{F_1 \sin \theta}{F_2 + F_1 \cos \theta} \right)$$



Special case $\rightarrow$

if $\theta = 90^\circ$, then

$$R = \sqrt{F_1^2 + F_2^2}$$

$$\angle \beta = \tan^{-1} \left(\frac{F_1}{F_2} \right)$$

if F_1 is vertical (F_v), then F_2 will be Horizontal (F_H).

So, $R = \sqrt{F_H^2 + F_v^2}$

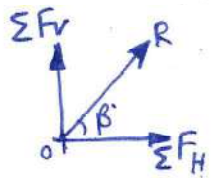
$$\angle \beta = \tan^{-1} \left(\frac{F_v}{F_H} \right)$$

Now in case of reversed forces, each and every force is resolved into two components — one Horizontal and one vertical. Then, horizontal forces are added up as per their direction (left/right), and vertical forces are also added up as per their direction (up/down). Let, these final Horizontal and Vertical forces are ΣF_H and ΣF_v respectively. Then,

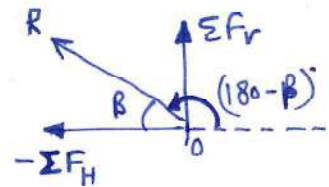
$$R = \sqrt{(\Sigma F_H)^2 + (\Sigma F_v)^2}$$

$$\angle \beta = \tan^{-1} \left(\frac{\Sigma F_v}{\Sigma F_H} \right)$$

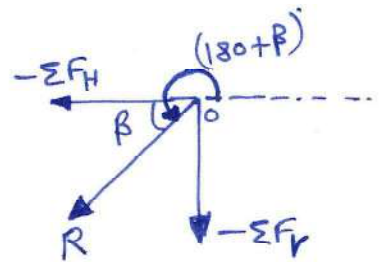
Case-I, if ΣF_v and ΣF_H both are +ve, then
actual angle of resultant = β



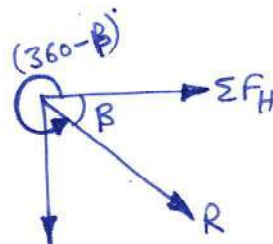
Case-II, if ΣF_v is +ve, but ΣF_H is negative, then
actual angle of resultant = $(180 - \beta)$



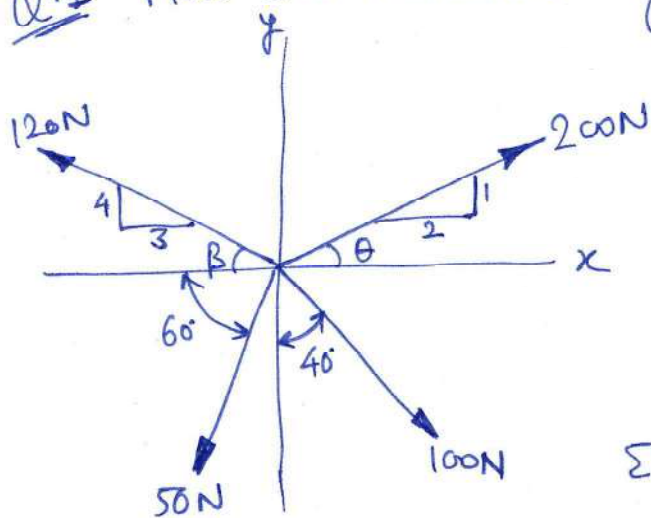
Case-III, if ΣF_v and ΣF_H both are -ve, then
actual angle of resultant = $(180 + \beta)$



Case-IV, if ΣF_v is -ve but ΣF_H is +ve, then
actual angle of resultant = $(360 - \beta)$
or $(-\beta)$



Q.1 Find the resultant magnitude and direction —



Sol:

$$\frac{\sqrt{5}}{2} \Rightarrow \sin \theta = \frac{1}{\sqrt{5}}$$

$$\cos \theta = \frac{2}{\sqrt{5}}$$

$$\frac{4}{3} \Rightarrow \sin \beta = \frac{4}{5}, \cos \beta = \frac{3}{5}$$

$$\Sigma F_x = 200 \cos \theta + 100 \sin 40^\circ - 120 \cos \beta - 50 \cos 60^\circ$$

$$= 200 \times \frac{2}{\sqrt{5}} + 100 \sin 40^\circ - 50 \cos 60^\circ - 120 \times \frac{3}{5}$$

$$= 146.164 \quad (+ve)$$

$$\Sigma F_y = 200 \sin \theta - 100 \cos 40^\circ + 120 \sin \beta - 50 \sin 60^\circ$$

$$= 200 \times \frac{1}{\sqrt{5}} - 100 \cos 40^\circ + 120 \times \frac{4}{5} - 50 \sin 60^\circ$$

$$= 65.537 \quad (+ve)$$

$$\Rightarrow \text{Resultant } R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$$

$$= \sqrt{(146.164)^2 + (65.537)^2}$$

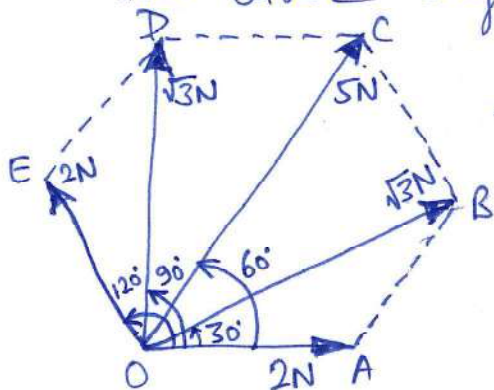
$$\boxed{R = 160.184 \text{ N}} \text{ Ans.}$$

Angle of resultant with +ve x-axis:

$$\theta_R = \tan^{-1} \left(\frac{\Sigma F_y}{\Sigma F_x} \right) = \tan^{-1} \left(\frac{65.537}{146.164} \right)$$

$$\boxed{\theta_R = 24.15^\circ} \text{ Ans.}$$

Q.2 Find the resultant of forces 2, $\sqrt{3}$, 5, $\sqrt{3}$ and 2 N that act at an angular point of a regular hexagon towards the other angular points taken in order.



Sol.

$\angle OAB = 120^\circ$ (hexagon angle)

Sides $OA = AB$

$\Rightarrow \angle AOB = \angle ABO = \boxed{30^\circ}$ each.

$\angle OCB = \frac{120^\circ}{2} = 60^\circ$ (symmetrical diagonal OC)

$\angle OBC = 120 - 30 = 90^\circ$

$\Rightarrow \angle BOC = 180 - (60 + 90) = 30^\circ$

$\Rightarrow \angle AOC = 30 + 30 = \boxed{60^\circ}$

Similarly, $\angle DOC = 30^\circ = \angle EOD$

$\Rightarrow \angle AOD = 60 + 30 = \boxed{90^\circ}$

and $\angle AOE = 90 + 30 = \boxed{120^\circ}$

$$\Sigma F_x = 2 + \sqrt{3} \cos 30^\circ + 5 \cos 60^\circ + \sqrt{3} \cos 90^\circ + 2 \cos 120^\circ$$

$$= 5 \text{ N}$$

$$\Sigma F_y = \sqrt{3} \sin 30^\circ + 5 \sin 60^\circ + \sqrt{3} \sin 90^\circ + 2 \sin 120^\circ$$

$$= 5\sqrt{3} \text{ N}$$

$$\Rightarrow R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2} = \sqrt{5^2 + (5\sqrt{3})^2} = \sqrt{100} \\ = \boxed{10\text{N}} \text{ Ans.}$$

$$\text{Angle of the resultant } \theta_R = \tan^{-1} \left(\frac{\Sigma F_y}{\Sigma F_x} \right) = \tan^{-1} \left(\frac{5\sqrt{3}}{5} \right) \\ = \tan^{-1}(\sqrt{3}) = \boxed{60^\circ} \text{ Ans.}$$

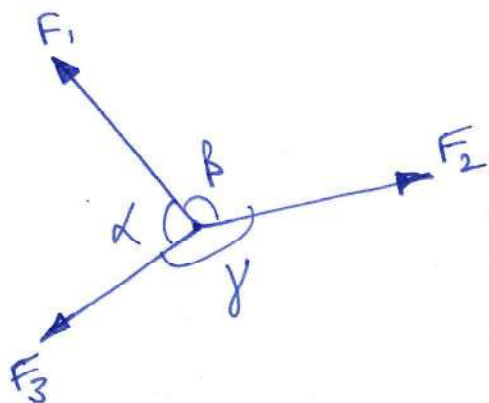
⑩ Condition of static equilibrium in case of a coplaner, concurrent force system -

static equilibrium means, the resultant of all the forces must be zero. For this, following conditions are necessary -

$$\text{and } \boxed{\begin{matrix} \Sigma F_x = 0; \\ \Sigma F_y = 0 \end{matrix}}$$

These two mathematical conditions provide us two equations, with the help of which we can find two known variables. In case of a coplaner, concurrent 3-force system, we can use "Lami's theorem".

⑪ LAMI's Theorem: "If three forces are concurrent, and are in static equilibrium, then the ratio of a force magnitude, the 'sine' of the angle between other two forces remains constant."



$$\boxed{R=0} \text{ (Static eqn)}$$

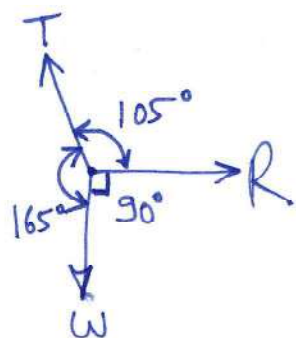
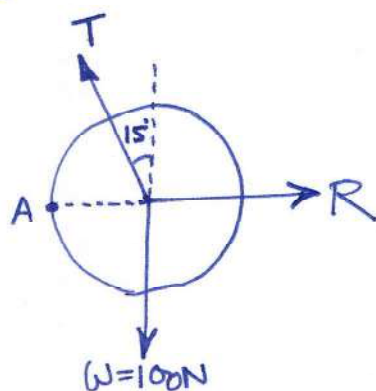
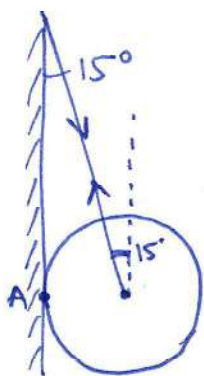
then,

$$\boxed{\frac{F_1}{\sin \gamma} = \frac{F_2}{\sin \alpha} = \frac{F_3}{\sin \beta} = \text{Constant}}$$

Q.3 A sphere of weight 100N is tied to a smooth wall by a string. Find the tension in the string and Reaction of the wall.

Sol.

Drawing the FBD of the sphere—



$$\text{Angle between } T \text{ and } R = 90 + 15 = 105^\circ$$

$$\text{Angle between } T \text{ and } 100N = 180 - 15 = 165^\circ$$

$$\text{Angle between } R \text{ and } 100N = 90^\circ$$

Applying Lami's theorem,

$$\frac{T}{\sin 90} = \frac{R}{\sin 165} = \frac{100}{\sin 105}$$

from 1st and 3rd,

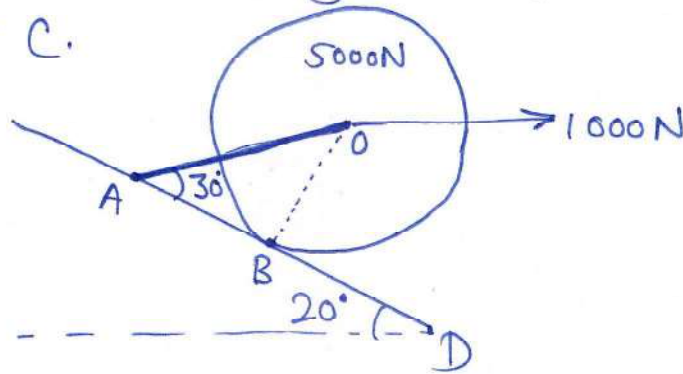
$$\boxed{T = 103.53N}$$

from 2nd and 3rd,

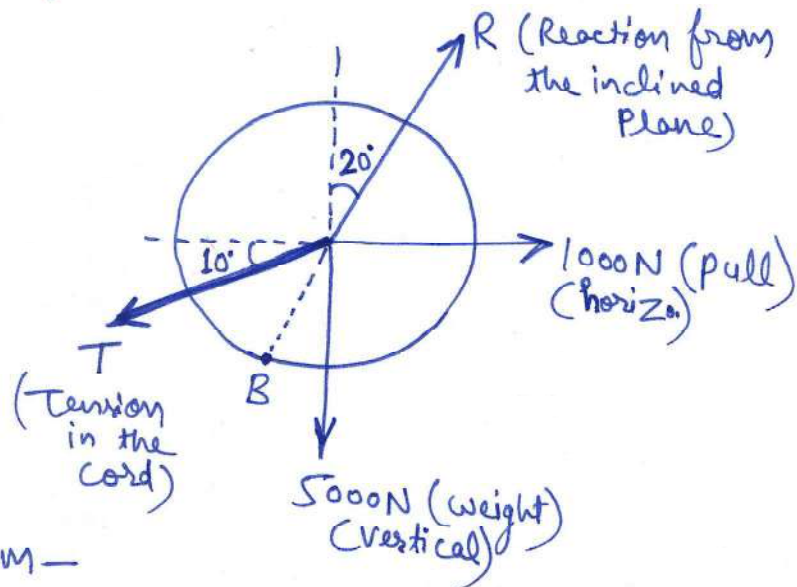
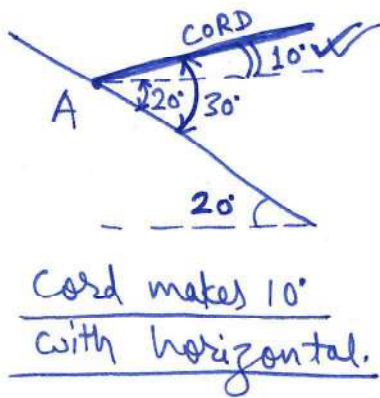
$$\boxed{R = 26.79N}$$

[Note: Such problems can also be solved using equilibrium equations. $\sum F_x = 0$ and $\sum F_y = 0$]

Q.4 A right circular roller of weight 5000N rests on a smooth inclined plane and is held in position by a cord AC as shown in the figure. Find the tension in the cord if there is a horizontal force of magnitude 1000N acting at C.



Sol. Drawing the FBD of the roller—



Conditions of equilibrium—

$$\begin{aligned} \Sigma F_x = 0, \quad & \{ 1000 + R \sin 20^\circ - T \cos 10^\circ = 0 \} \quad \times \cos 20^\circ \\ \Sigma F_y = 0, \quad & \{ R \cos 20^\circ - 5000 - T \sin 10^\circ = 0 \} \quad \times \sin 20^\circ \end{aligned}$$

$$\Rightarrow 1000 \cos 20^\circ + R \sin 20^\circ \cos 20^\circ - T \cos 10^\circ \cos 20^\circ = 0 \quad \text{--- (I)}$$

$$\text{and } -5000 \sin 20^\circ + R \sin 20^\circ \cos 20^\circ - T \sin 10^\circ \sin 20^\circ = 0 \quad \text{--- (II)}$$

Subtracting,

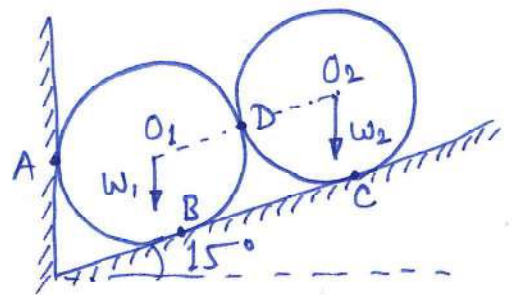
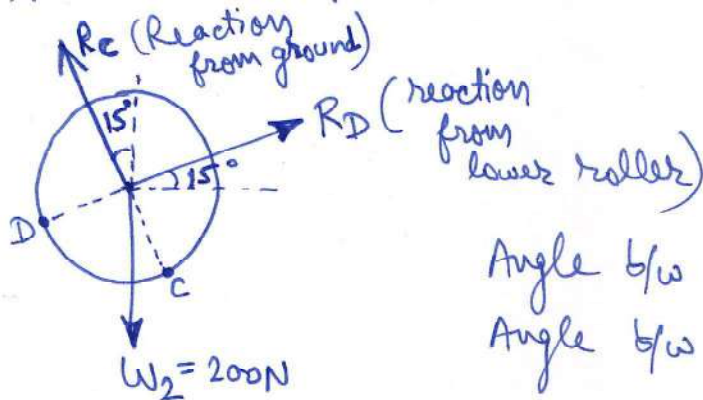
$$2649.7933 - 0.8660T = 0$$

$$\Rightarrow T = \frac{2649.7933}{0.8660} = \boxed{3059.8 \text{ N}} \quad \text{Ans.}$$

$$\Rightarrow \boxed{R = 5886.31 \text{ N}} \quad \underline{\text{Ans.}}$$

Q.5 Two rollers of the same diameter are supported by an inclined plane and a vertical wall as shown in the figure. The upper and the lower rollers are respectively 200N and 250N in weight. All the surfaces are smooth. Calculate the reactions at all the points of contact.

Sol. Drawing the FBD of the upper roller first—



Angle b/w R_c and $R_d = 90^\circ$

Angle b/w R_d and $200\text{N} = 90 + 15$
 $= 105^\circ$

Angle betw 200N and $R_c = 180 - 15$
 $= 165^\circ$

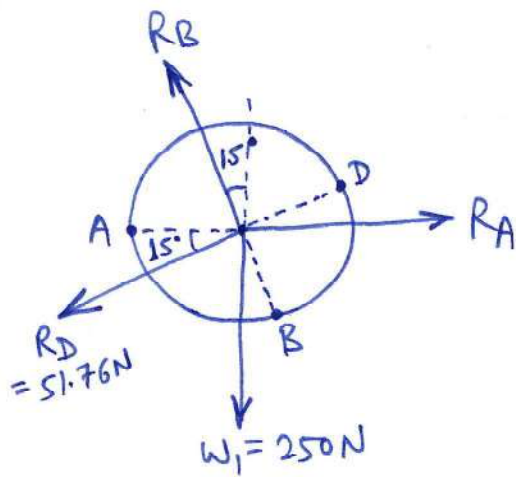
$\Rightarrow$ Applying Lami's Theorem—

$$\frac{R_d}{\sin 165^\circ} = \frac{R_c}{\sin 105^\circ} = \frac{200\text{N}}{\sin 90^\circ}$$

$$\Rightarrow \boxed{R_d = 51.76 \text{ N}} \quad \underline{\text{Ans}} \quad (\text{Using 1st and 3rd part})$$

$$\text{and } \boxed{R_c = 193.18 \text{ N}} \quad \underline{\text{Ans}} \quad (\text{Using 2nd and 3rd part})$$

Now R_d is known. We will use it in FBD of the lower roller. Drawing the FBD of the lower roller—



$$\Sigma F_x = 0,$$

$$\Rightarrow R_A - 51.76 \cos 15^\circ - R_B \sin 15^\circ = 0 \quad \text{--- (I)}$$

$$\Sigma F_y = 0,$$

$$\Rightarrow R_B \cos 15^\circ - 51.76 \sin 15^\circ - 250 = 0 \quad \text{--- (II)}$$

$$\Rightarrow R_B \cos 15^\circ = 263.3965$$

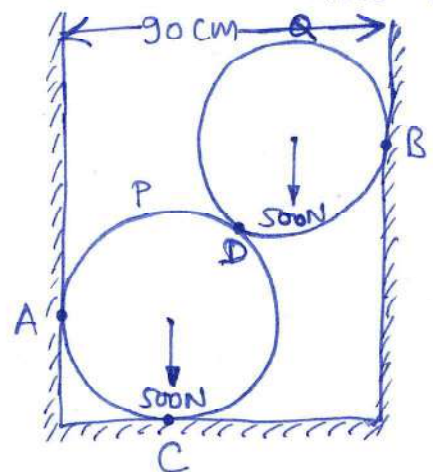
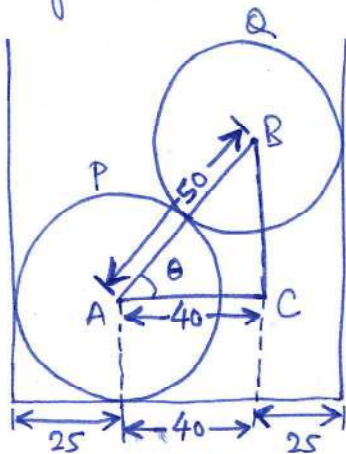
$$\Rightarrow \boxed{R_B = 272.688 \text{ N}} \quad \text{Ans.}$$

putting this value in any of equation (I),

$$\boxed{R_A = 120.57 \text{ N}} \quad \text{Ans.}$$

Q.6: Two smooth spheres P, Q each of radius 25 cm and weight 500 N rest in a horizontal channel having vertical walls. If the distance between the walls is 90 cm, make calculations for the force exerted on the wall and floor.

Sol.



(above diagram is given)

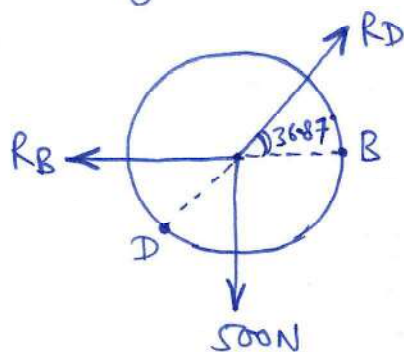
Considering ΔABC ,

$$AC = 90 - (25 + 25) = 40 \text{ cm.}$$

$$AB = 25 + 25 = 50 \text{ cm.}$$

$$\Rightarrow \cancel{\tan \theta} \cos \theta = \frac{40}{50} \Rightarrow \theta = \underline{\underline{36.87^\circ}}$$

Drawing the FBD of upper roller —



Applying Lami's theorem —

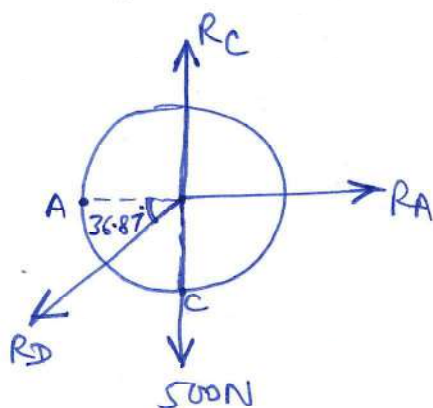
$$\frac{R_B}{\sin(90 + 36.87)} = \frac{R_D}{\sin 90} = \frac{500}{\sin(180 - 36.87)}$$

$$\Rightarrow R_D = 833.33 \text{ N}$$

$$\text{and } R_B = 833.33 \cdot \sin 126.87^\circ$$

$$R_B = 666.66 \text{ N} \quad \underline{\text{Ans.}}$$

Drawing FBD of the lower roller —



$$\sum F_x = 0,$$

$$R_A - R_D \cos 36.87 = 0$$

$$\text{or } R_A = 833.33 \cos 36.87$$

$$R_A = 666.66 \text{ N} \quad \underline{\text{Ans.}}$$

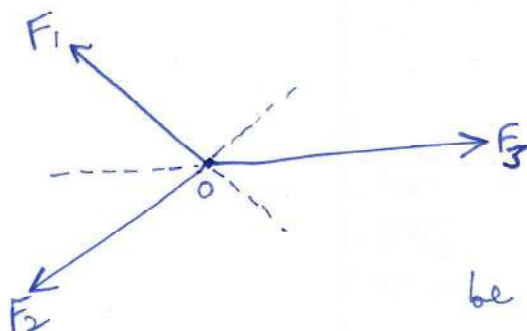
$$\sum F_y = 0,$$

$$R_C - 500 - R_D \sin 36.87^\circ = 0$$

$$R_C = 500 + 833.33 \sin 36.87$$

$$R_C = 1000 \text{ N} \quad \underline{\text{Ans.}}$$

Condition of concurrency — "If three forces are in static equilibrium, then the lines of action of these forces must pass through a single point."

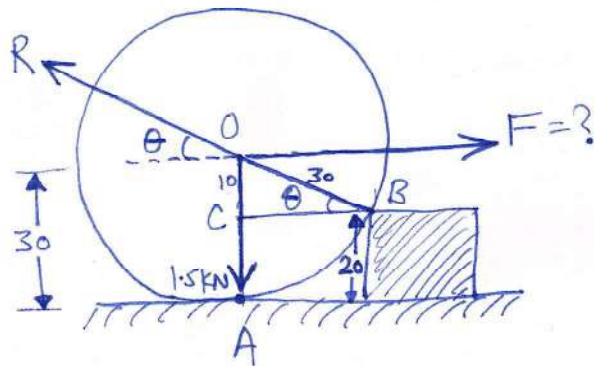
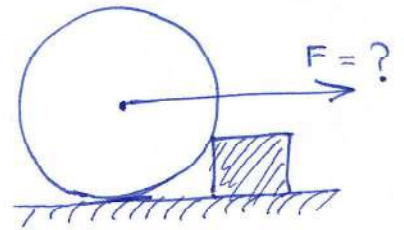


here, F_1 , F_2 and F_3 are in static equilibrium.

Therefore, the resultant of any two forces must be equal and opposite to the line of action of third force.

Q. A wheel of 60 cm diameter and 1.5 kN weight rests against a rectangular obstacle of height 20 cm. Find the least horizontal pull to be applied (a) Through the centre of the wheel, (b) through the top point on its vertical diameter, to over-turn the wheel over the obstacle?

Sol. (a) In case of the minimum pull is applied to just over-turn the wheel, it just leaves the ground and lifts up. Therefore, the ground reaction becomes zero. Hence, it becomes a three-force system. Let us draw the FBD —



In $\triangle OCB$, $OC = 30 - 20 = 10$ cm.

$OB = 30$ cm. (radius)

$$\Rightarrow \sin \theta = \frac{10}{30} \text{ or } \theta = \sin^{-1}\left(\frac{10}{30}\right)$$

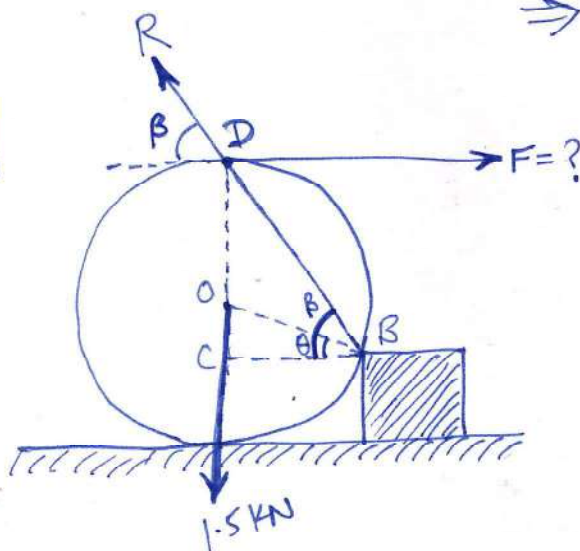
$$\theta = 19.47^\circ$$

Applying Lami's th^m $\rightarrow$

$$\frac{R}{\sin 90} = \frac{F}{\sin(90 + 19.47)} = \frac{1.5}{\sin(180 - 19.47)}$$

$$\Rightarrow R = 4.5 \text{ kN and } \boxed{F = 4.243 \text{ kN}} \text{ Ans.}$$

(b)



In right angled $\triangle OCB$,

$$CB = \sqrt{OB^2 - OC^2} = \sqrt{30^2 - 10^2}$$

$$= 28.284 \text{ cm.}$$

$$\text{and } DC = DO + OC$$

$$= 30 + 10 = 40 \text{ cm}$$

$\Rightarrow \Delta DCB,$

$$\tan \beta = \frac{DC}{CB} = \frac{40}{28.284} \Rightarrow \beta = 54.735^\circ$$

Applying Lami's theorem at point D, which is the point of concurrency of all the three forces in equilibrium (ignoring the ground reaction):

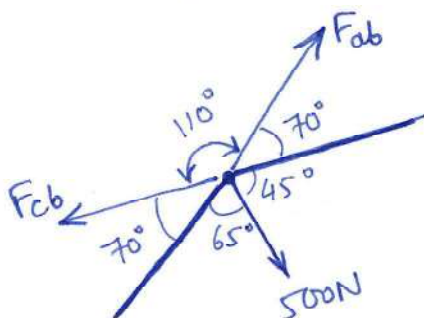
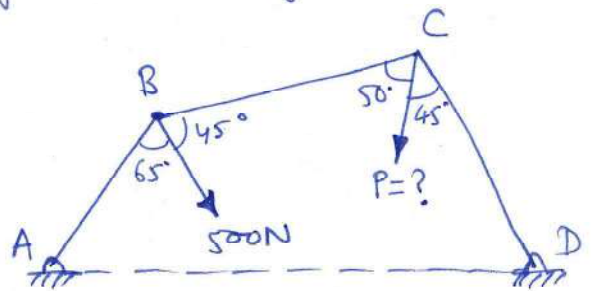
$$\frac{R}{\sin 90} = \frac{F}{\sin (90 + 54.735)} = \frac{1.5 \text{ kN}}{\sin (180 - 54.735)}$$

$$R = 1.837 \text{ kN}$$

$$\text{and } \boxed{F = 1.060 \text{ kN}} \text{ Ans.}$$

Q. Three bars hinged at A and D and pinned at B and C shown in the figure below form a four link mechanism. Determine the value of P that will prevent movement of bars. Also calculate the force transferred in bars AB and CD

Sol. Any force which transfers through the bar must be along its length. Analyzing the joint B.



Let, Force on B from A = F_{AB}
and force on B from C = F_{CB}

$$\frac{500}{\sin 110} = \frac{F_{AB}}{\sin 135} = \frac{F_{CB}}{\sin 115}$$

$$\Rightarrow F_{AB} = \frac{500 \times \sin 135}{\sin 110} = \boxed{376.24 \text{ N}} \text{ Ans.}$$

$$\text{also, } \frac{500}{\sin 110^\circ} = \frac{F_{cb}}{\sin 115^\circ}$$

$$\Rightarrow F_{cb} = \frac{500 \times \sin 115^\circ}{\sin 110^\circ} = \boxed{482.23 \text{ N}}$$

Ans.

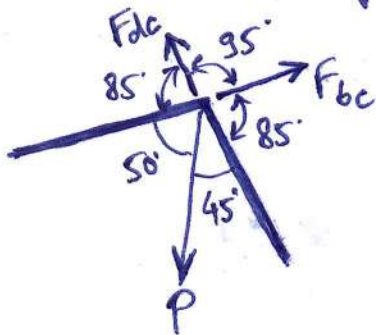
Now,

Analyzing junction C

Let, force from B on C is F_{bc}

$$\text{and } F_{bc} = F_{cb} = 482.23 \text{ N}$$

and force from D on C is F_{dc} .



Applying Lami's theorem —

$$\frac{F_{bc}}{\sin 135^\circ} = \frac{P}{\sin 95^\circ} = \frac{F_{dc}}{\sin 130^\circ}$$

Using 1st and 2nd parts,

$$\frac{482.23}{\sin 135^\circ} = \frac{P}{\sin 95^\circ} \Rightarrow \boxed{P = 679.38 \text{ N}}$$

Ans.

Using 1st and 3rd parts of above equation,

$$\frac{482.23 \text{ N}}{\sin 135^\circ} = \frac{F_{dc}}{\sin 130^\circ}$$

$$\Rightarrow \boxed{F_{dc} = 522.42 \text{ N}}$$

Ans.

(12) Beams and its types

A beam is a long, straight structural element that supports loads applied along its length. Specifically, beams are designed to withstand transverse loads. It has one or more supports.

Types of loads on a Beam: 1) Point load or concentrated load

2) Uniformly Distributed load.
(UDL)

3) Uniformly Varying load
(UVL)

4) Concentrated moment or point-moment

1) Point load — A transverse load acting on a single point, measured in terms of N or kN .



2) UDL — A load that is spread evenly across a length or area. It is defined in terms of load intensity, in N/m or kN/m .

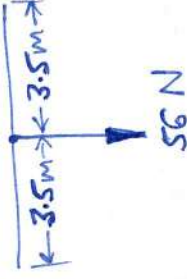


OR $\frac{x N/m}{\text{length}} \rightarrow \text{total load} = x y N$

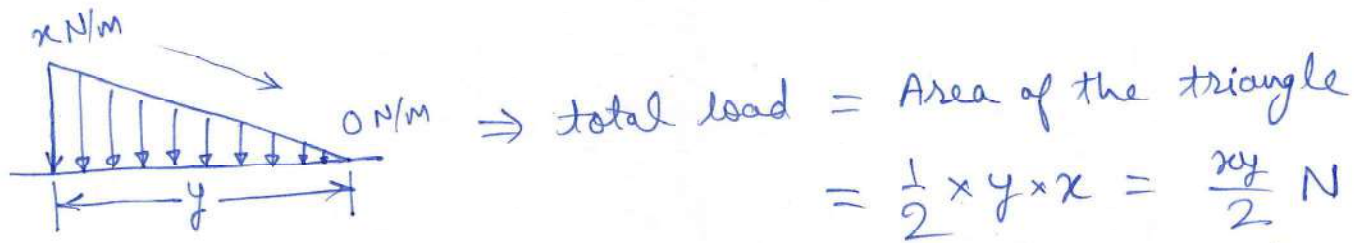
$= \text{load intensity} \times \text{length.}$

the total load of UDL can be assumed to act on the mid point of its length.

example,

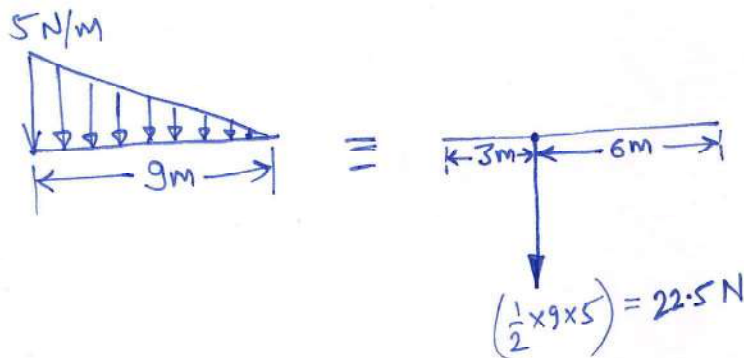
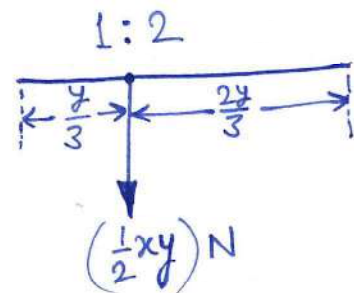
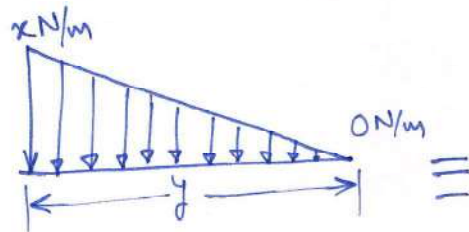


- 3) UVL - A load that increases or decreases linearly across a structure, such as a beam. It can also be considered as a triangular load. The intensity of load (in N/m) changes uniformly from a minimum value to a maximum value.

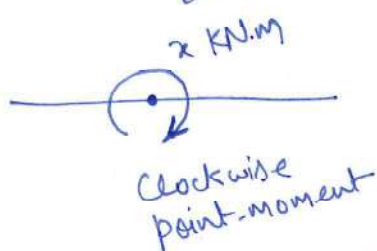
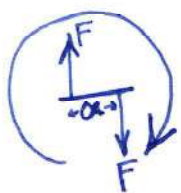


this total load can be assumed to act at a point, which divides the length in the ratio of 1:2 (1 on right angle side).

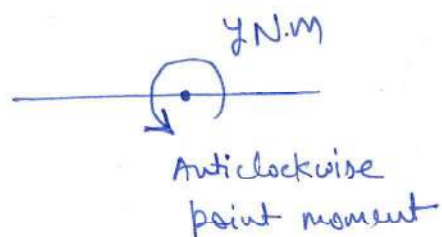
example,



- 4) Point moment - As the name suggests, it is not a force. It is a combination of 2 equal and opposite forces acting at a gap, which causes pure turning effect.



or

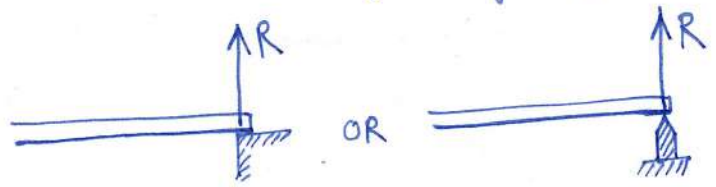


It is also mentioned as "Couple".

It is measured in Nm or Nmm or KN.m, etc.

⑬ Types of Supports — Beam supports transfer the forces and loads from a beam to the other parts of the structure. There are various types of supports —

1) Simple support —
or
Knife edge support —



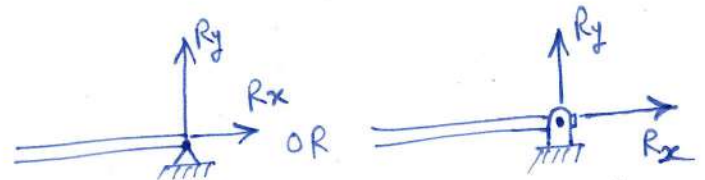
It causes reaction only in one direction, $\perp$ to the beam.

2) Roller Support —



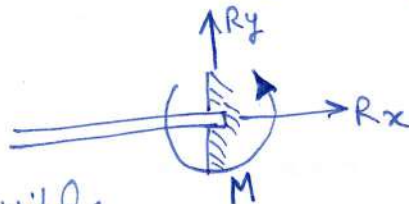
It allows axis movement in the beam, and rotation of the beam is also allowed. It also causes only one reaction, $\perp$ to the base of support.

3) Hinge Support or pin joint —



It allows only rotation of the beam, and restricts its linear movements. It will cause two reactions — one horizontal and one vertical (one of them may be zero too).

4) Fixed Support —



It stops all the possible movements of the beam (linear or rotational). It causes one horizontal, one vertical ~~one~~ reaction and one point moment (reaction moment) at the support.

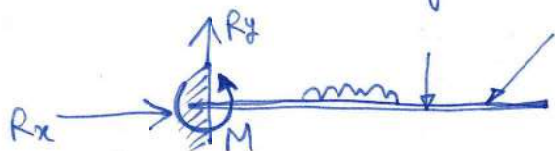
④ Types of beams —

- 1) Simply supported Beam — It is a beam that rests on two supports and is free to move (or expand) along its length. It has pin joint on one end and roller support (or simple support) on the other end.



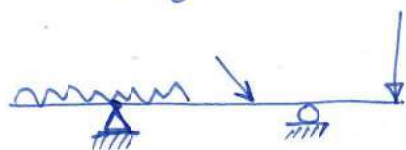
{ max 3 unknown reactions }

- 2) Cantilever Beam — It has on one side fixed support and other side is free.

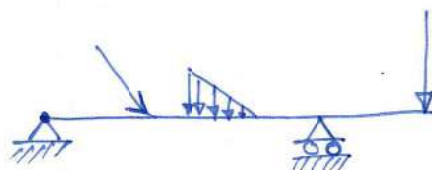


{ Max. 3 unknowns }

- 3) Over hanging beam — When in a simply supported beam, the loads are acting beyond the support-positions also. It may be one side or both side over-hanging beam.

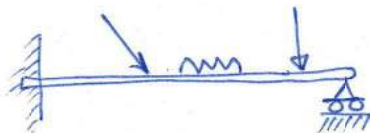


Both side over hanging

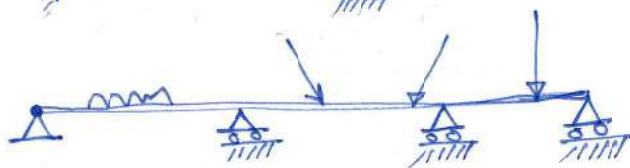


One side over hanging.

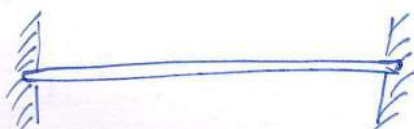
- 4) Propped Cantilever



- 5) Continuous Beams



- 6) Both side Fixed



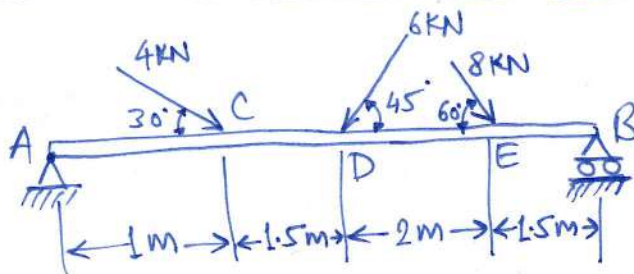
Various forces which act on a beam come under the category of coplanar - non-concurrent force system. For a system of coplanar, non-concurrent force system, the required conditions of static equilibrium are —

$$\left. \begin{aligned} \Sigma F_x &= 0, \\ \Sigma F_y &= 0 \\ \text{and } \Sigma M &= 0 \text{ (about any point)} \end{aligned} \right\}$$

Using the above equations, we can find the unknown reactions acting on any type of loaded beam, in which there are a maximum three unknown variables.

15 Problems Based on Calculation of Support Reactions

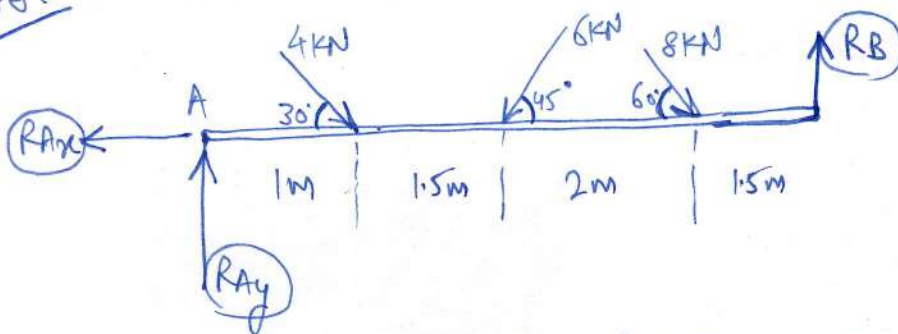
Q.



Calculate the support reactions.

Sol.

Let us show the reactions at the places of supports —



$$\Sigma F_y = 0, \quad R_{Ay} + R_B - 4 \sin 30^\circ - 6 \sin 45^\circ - 8 \sin 60^\circ = 0$$

$$\Rightarrow R_{Ay} + R_B = 13.17 \quad \text{--- (1)}$$

$$\Sigma M_A = 0, \quad (4 \sin 30^\circ) \times 1 + (6 \sin 45^\circ) \times 2.5 + (8 \sin 60^\circ) \times 4.5 - R_B \times 6 = 0$$

$$6R_B = 43.783$$

$$\Rightarrow \boxed{R_B = 7.297 \text{ kN}} \quad \text{Ans.}$$

from equation (1),

$$R_{Ay} + 7.297 = 13.17$$

$$\Rightarrow \boxed{R_{Ay} = 5.87 \text{ kN}}$$

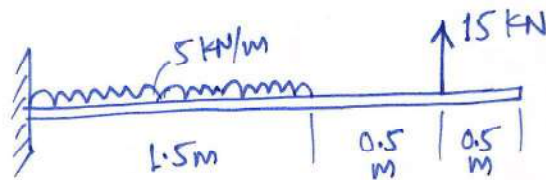
$$\Sigma F_x = 0,$$

$$4 \cos 30^\circ - 6 \cos 45^\circ + 8 \cos 60^\circ - R_{Ax} = 0$$

$$\Rightarrow \boxed{R_{Ax} = 3.22 \text{ kN}} \text{ (left side)} \leftarrow$$

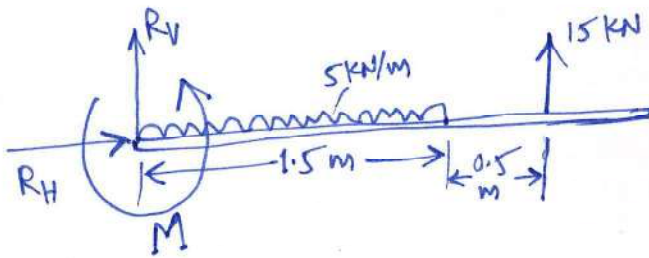
Note: In case if opposite direction of R_{Ax} was assumed, then the magnitude of the answer will be same (3.22 kN), and only the direction (i.e. sign) will be opposite.

Q.



Calculate the support reactions for the given cantilever beam.

Sol.



$$\Sigma F_y = 0, \Rightarrow R_V + 15 - \underbrace{5 \times 1.5}_{\text{UDL}} = 0$$

$$R_V + 7.5 = 0 \Rightarrow R_V = -7.5 \text{ kN}$$

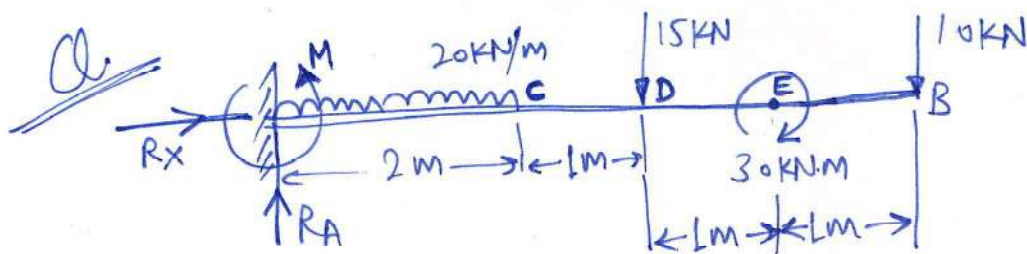
or $\boxed{R_V = 7.5 \text{ kN downward}}$
(opposite to the assumed direction)

$$\Sigma F_x = 0 \Rightarrow \boxed{R_H = 0}$$

$$\Sigma M_A = 0$$

$$-M + \underbrace{7.5 \times 0.75}_{\text{moment of UDL}} - 15 \times 2 = 0 \Rightarrow M = -24.375$$

or $\boxed{M = 24.375 \text{ kN.m Clockwise}}$
(again, opposite to the assumed direction)



Calculate the support reactions.

Sol. Assumed directions of the support reactions at the fixed end are already shown.

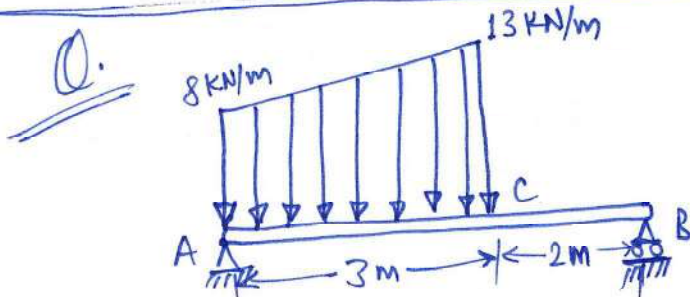
$$\underline{\Sigma M_A = 0} \Rightarrow -M + \underbrace{(20 \times 2) \times 1}_{\text{moment of UDL}} + 15 \times 3 + \underbrace{30}_{\text{point moment}} + 10 \times 5 = 0$$

$$\Rightarrow \boxed{M = 165 \text{ kN}\cdot\text{m}} \text{ Ans.}$$

$$\underline{\Sigma F_x = 0} \Rightarrow \boxed{R_x = 0}$$

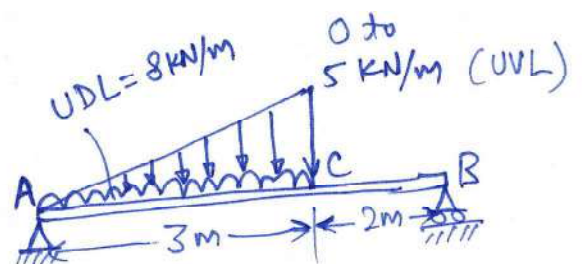
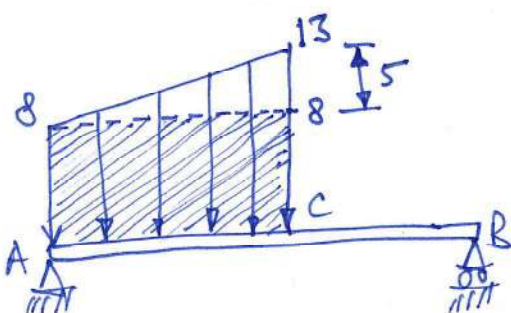
$$\underline{\Sigma F_y = 0} \Rightarrow R_A - \underbrace{(20 \times 2)}_{\text{UDL}} - 15 - 10 = 0$$

$$\boxed{R_A = 65 \text{ kN}} \text{ Ans.}$$



Calculate the support reactions.

Sol. The given load is also a UVL, but it is varying from 8 kN/m to 13 kN/m linearly. It can be broken into two overlapping loads — 1 UDL + 1 UVL.



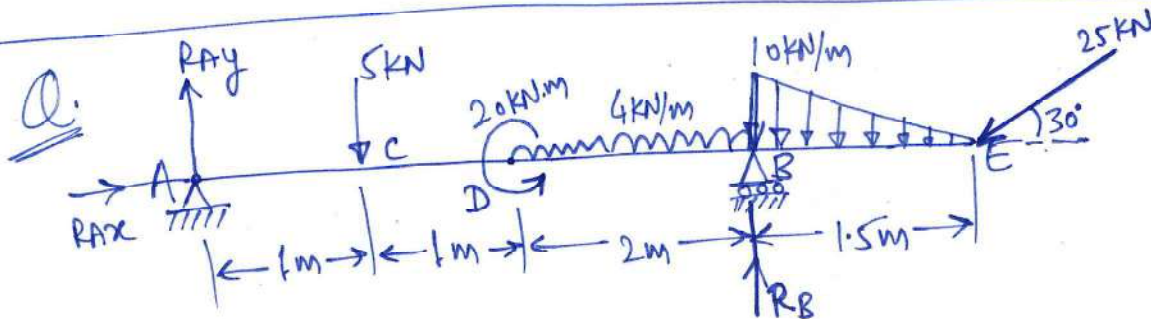
$$\underline{\Sigma M_A = 0} \Rightarrow \underbrace{\left(\frac{1}{2} \times 3 \times 5\right) \times 2}_{\text{moment of UVL}} + \underbrace{(8 \times 3) \times 1.5}_{\text{moment of UDL}} - R_B \times 5 = 0$$

$$\Rightarrow \boxed{R_B = 10.2 \text{ kN}} \text{ Ans.}$$

$$\underline{\Sigma F_y = 0} \quad R_A + R_B - \underbrace{\left(\frac{1}{2} \times 3 \times 5\right)}_{\text{UVL}} - \underbrace{(8 \times 3)}_{\text{UDL}} = 0$$

$$R_A + R_B = 31.5$$

$$R_A + 10.2 = 31.5 \Rightarrow \boxed{R_A = 21.3 \text{ kN}} \text{ Ans.}$$



Calculate the support reactions.

Sol.

$$\underline{\Sigma F_x = 0} \Rightarrow R_{Ax} - 25 \cos 30^\circ = 0 \Rightarrow \boxed{R_{Ax} = 21.65 \text{ kN}} \text{ Ans.}$$

$$\underline{\Sigma F_y = 0} \Rightarrow R_{Ay} + R_B - 5 - (4 \times 2) - \left(\frac{1}{2} \times 1.5 \times 10\right) - 25 \sin 30^\circ = 0$$

$$R_{Ay} + R_B = 33 \quad \text{--- (1)}$$

$$\underline{\Sigma M_A = 0} \Rightarrow 5 \times 1 - 20 + (4 \times 2) \times 3 + \left(\frac{1}{2} \times 1.5 \times 10\right) \times 4.5 - R_B \times 4 + 25 \sin 30^\circ \times 5.5 = 0$$

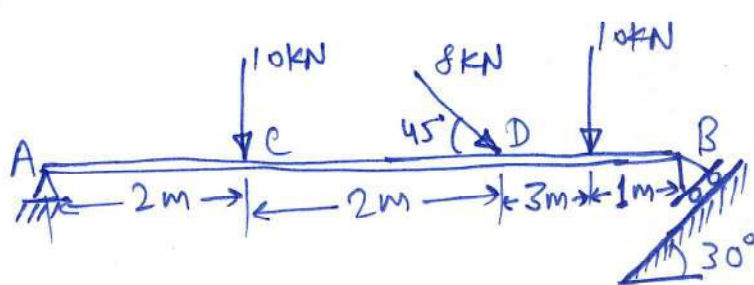
$$\Rightarrow \boxed{R_B = 27.875 \text{ kN}}$$

putting this value in eqn (1), we get

$$R_{Ay} = 33 - R_B$$

$$= 33 - 27.875 = \boxed{5.125 \text{ kN}} \text{ Ans.}$$

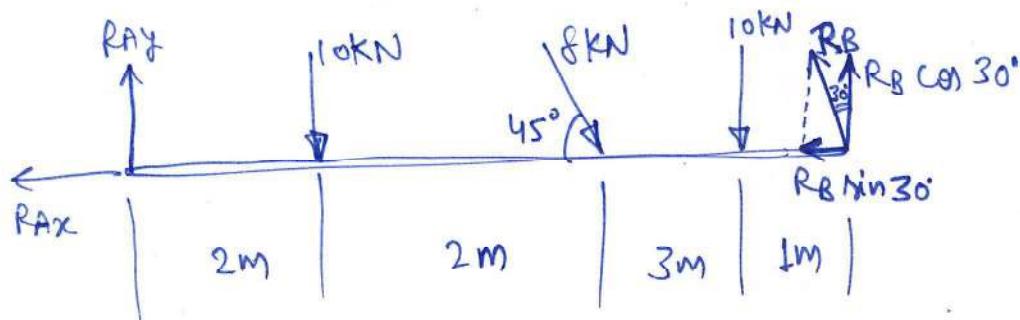
Q.



Calculate the support reactions.

Sol.

When the base of the roller support is inclined, the reaction will also be inclined, and will be $\perp$ to the inclined plane. So, drawing the FBD -



$$\underline{\Sigma F_x = 0} \quad -R_{ax} + 8 \cos 45^\circ - R_B \sin 30^\circ = 0 \quad \text{--- (I)}$$

$$\underline{\Sigma F_y = 0} \quad R_{ay} - 10 - 8 \sin 45^\circ - 10 + R_B \cos 30^\circ = 0 \quad \text{--- (II)}$$

$$\underline{\Sigma M_A = 0} \quad 10 \times 2 + 8 \sin 45^\circ \times 4 + 10 \times 7 - R_B \cos 30^\circ \times 8 = 0$$

$$112.627 = 8 R_B \cos 30^\circ$$

$$\Rightarrow \boxed{R_B = 16.25 \text{ KN}} \quad \underline{\text{Ans.}}$$

putting this value in eqⁿ (I) -

$$R_{ax} = 8 \cos 45^\circ - 16.25 \sin 30^\circ$$

$$\Rightarrow R_{ax} = -2.468 \text{ KN}$$

$$\text{or } \boxed{R_{ax} = 2.468 \text{ KN towards right hand side.}}$$

from eqⁿ (II),

$$\boxed{R_{ay} = 11.583 \text{ KN}} \quad \underline{\text{Ans.}}$$

Unit-1

Basics of Mechanical and Automation Engineering (BMA)

Contents: PART-2

Strength of Materials

- ① Stress
 - ② Strain
 - ③ Normal Stress (longitudinal stress)
 - ④ Normal strain (longitudinal strain)
 - ⑤ Tangential Loads
 - ⑥ Shear Stress
 - ⑦ Shear strain
 - ⑧ Volumetric Stress
 - ⑨ Volumetric Strain
 - ⑩ Hooke's Law
 - ⑪ Young's Modulus, Modulus of Rigidity and Bulk Modulus
 - ⑫ Lateral Strain
 - ⑬ Poisson's Ratio
 - ⑭ Relation Between volumetric Strain and longitudinal strains
 ↳ (a) For one dimensional loading
 (b) For three dimensional loading
 - ⑮ Relation Between Young's Modulus (E) and Bulk Modulus (K)
 - ⑯ Relation Between Young's Modulus (E) and Mod. of Rigidity (C)
 - ⑰ Relation between E, C and K.
 - ⑱ Numerical Problems
 - ⑲ Stress-Strain diagram for mild steel and other ductile materials.
 - ⑳ Stress-Strain diagram for brittle materials.
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- ㉑ Unit Summary and Important Formulas
 - ㉒ Important 'Short Answer Type' questions.
 - ㉓ Important 'Long Answer Type' questions.
 - ㉔ Unit-1: Question Bank

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① Stress:

Solids have a property of resisting the externally applied forces, or any other type of external load. It is due to a fixed molecular arrangement. Internal forces are developed within the body which oppose deformation.

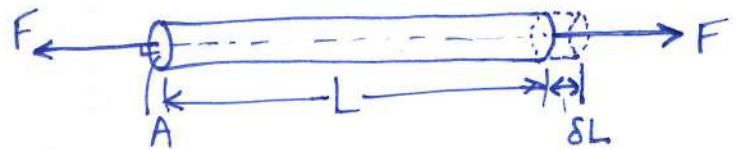
"The internal resistance (restoring force) developed in a body per unit area is known as stress."

② Strain:

It is defined as the ratio of change in dimension of an object, to the initial dimension. It has no unit.

③ Normal Stress:

In case of an axially loaded straight member,



the applied load is perpendicular to the cross-section area (i.e., along the length). The stress developed along the length is 'Normal stress' or 'longitudinal stress' or 'Axial stress'.

$$\boxed{\sigma = \frac{F}{A}} \quad \text{Unit: } \frac{\text{N}}{\text{m}^2} \quad \left[\text{Note: In case of equilibrium, applied load} = \text{restoring force} \right]$$

$$\frac{\text{N}}{\text{m}^2} = \text{Pascal (Pa)}, \quad 10^3 \text{ Pa} = \text{KPa}, \quad 10^6 \text{ Pa} = \text{MPa}, \quad 10^9 \text{ Pa} = \text{GPa}.$$

$$\text{also, } \boxed{1 \text{ N/mm}^2 = 1 \text{ MPa}}, \quad \boxed{1 \text{ kN/mm}^2 = 1 \text{ GPa}}$$

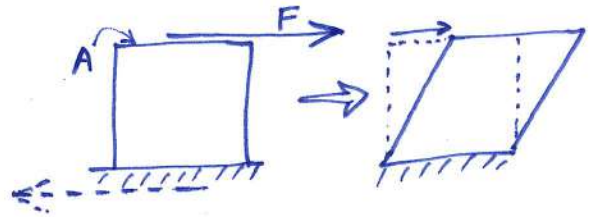
④ Normal strain: The ratio of change in length to the original (initial) length is called

'Normal strain' or 'longitudinal strain' or 'Axial strain'.

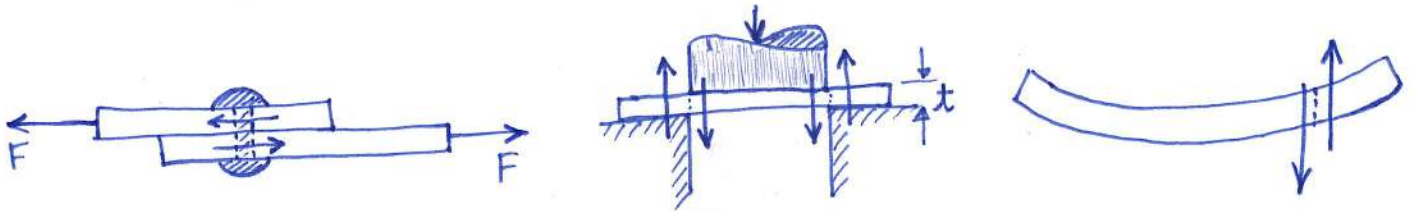
$$\epsilon = \frac{\delta L}{L} \quad (\text{Unitless})$$

③ Tangential loads: When the load is applied in tangential direction, it causes distortion in the shape of the object, i.e. the shape of the object changes.

It is always accompanied by equal and opposite force on the other side of the object.



The examples are pins, rivets, bolts, shears and scissors.



In case of beams, all transverse loads act tangential to the cross-section, and cause deflection. When two equal and opposite tangential forces act on two opposite sides of a single cross-section, they may cause 'shearing' or cutting of that piece (parting off). example- punching machine.

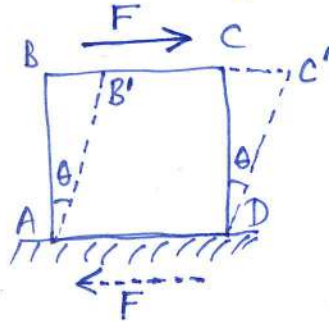
③ Shear stress:—

Tangential force acting on the surface of a body per unit surface area is called 'shear stress'.

$$\tau = \frac{\text{tangential force}}{\text{surface area}} = \frac{F}{A} \quad (\text{same unit, Pa}).$$

⑦ Shear Strain:

Shear loads cause sliding of a layer with respect to the adjacent layers. The layer on which the tangential load is applied experiences maximum shift. Let, the top layer BC shifts to another position B'C' due to the force F. Let, the gap between the top and bottom layer is AB or CD. Then,



$$\text{Shear strain} = \frac{BB'}{AB} \text{ or } \frac{CC'}{CD}$$

from the figure shown above, it is clear that

$$\frac{BB'}{AB} = \frac{CC'}{CD} = \tan \theta$$

if θ is very small, then $\tan \theta \approx \theta$

$$\Rightarrow \boxed{\text{Shear strain} = \theta}$$

$\Rightarrow$ Shear strain is the angle through which a face originally perpendicular to the fixed face is turned. It is the deformation of a material that occurs when parallel internal surfaces slide past each other due to shear stress.

⑧ Volumetric Stress:-

Also known as hydrostatic stress, is the stress which develops when deforming force acts from all directions, resulting in the change of volume

of the object. When an object is dipped inside a liquid like water, such type of stresses develop. Numerically, it is equal to the hydrostatic pressure p .

⑨ Volumetric Strain:

Due to the hydrostatic pressure, the volume of the object changes. The ratio of change in volume to the original volume of the object is known as Volumetric strain.

$$\text{Vol. Strain} = \frac{\delta V}{V}$$

⑩ Hooke's Law: It states that, "within the elastic limits, stress is directly proportional to the strain."

$$\sigma \propto E$$
$$\text{or } \boxed{\sigma = E \cdot \epsilon} \text{ or } \boxed{E = \frac{\sigma}{\epsilon}} \quad (\text{Unit: Pa, MPa, GPa...})$$

here, E is the constant of proportionality, which is known as "Modulus of elasticity". There are three different modulus of elasticity—

⑪ I) Young's Modulus = $\frac{\text{Longitudinal Stress}}{\text{Longitudinal strain}}$

$$E = \frac{F/A}{\delta L/L} = \frac{F \cdot L}{A \cdot \delta L}$$

$$\Rightarrow \boxed{\delta L = \frac{F \cdot L}{A E}}$$

Using this relation, we can find the elongation/contraction of a uniform ~~ex~~ cross-sectional bar under axial loads.

II) Mod. of Rigidity (C or G) = $\frac{\text{Shear stress}}{\text{shear strain}}$

$$C = \frac{\tau}{\theta}$$

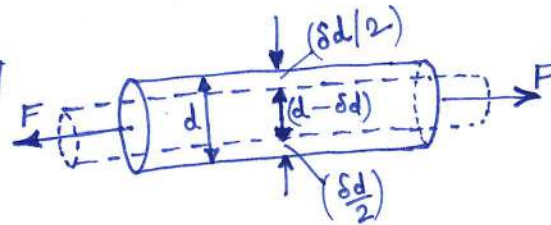
(if θ is small)
angular deformation)

III) Bulk modulus (K) = $\frac{\text{volumetric stress}}{\text{volumetric strain}}$

$$K = \frac{p}{(\Delta V/V)}$$

⑫ Lateral Strain: When a circular cross-sectional bar is subjected to normal tensile stress, it causes increase in length. But it also causes decrease in cross-section diameter. If the length changes from $(L + \delta L)$, and diameter changes from d to $(d - \delta d)$, then

$$\text{Lateral strain} = \frac{\delta d}{d}$$



"The strain which develops perpendicular to the direction of applied load is called lateral strain."

It means that if load is applied along x-axis, it will create longitudinal strain along x-axis, and lateral strains along y and z axes.

In case of square or rectangular cross-section, lateral strain is defined as $\frac{\text{Change in width}}{\text{original width}}$ ~~and~~ or $\frac{\text{Change in thickness}}{\text{original thickness}}$

$$\text{Lat. S.} = \frac{\delta w}{w} \text{ or } \frac{\delta t}{t}$$

(13) Poisson's Ratio (μ) :- It is the ratio of lateral strain to the longitudinal strain. It is a constant for each and every material. Its value lies in the range, 0 to 0.5. For most of the metals and alloys, its value lies between 0.2 to 0.45.

$$\mu = \frac{\text{lat. S.}}{\text{long. S.}} = \frac{\delta d/d}{\delta L/L}$$

or $\boxed{\text{lateral strain} = \mu \cdot \text{longitudinal strain}}$

Note: Poisson's ratio for a perfectly incompressible linear elastic material is 0.5.

(14) Relation between volumetric strain and longitudinal strains.

Let, the dimensions of an elastic block along x, y and z axes are l, b and h respectively. Then, Volume of the block $V = l \cdot b \cdot h$

taking log, $\log_e V = \log_e l + \log_e b + \log_e h$

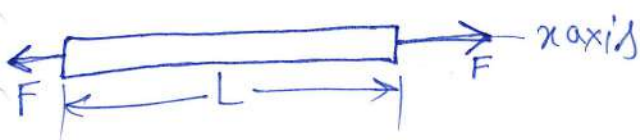
by differentiation,

$$\frac{1}{V} \cdot dV = \frac{1}{l} \cdot dl + \frac{1}{b} \cdot db + \frac{1}{h} \cdot dh$$

or Volumetric Strain = (strain along x-axis + strain along y axis + strain along z-axis.)

$$\boxed{\epsilon_v = \epsilon_x + \epsilon_y + \epsilon_z}$$

(a) In case of one-directional loading (along x-axis) -



$$\left. \begin{aligned} \epsilon_x &= \frac{\delta L}{L} \\ \epsilon_y &= -\mu \cdot \frac{\delta L}{L} \\ \epsilon_z &= -\mu \cdot \frac{\delta L}{L} \end{aligned} \right\} \text{both lateral strain}$$

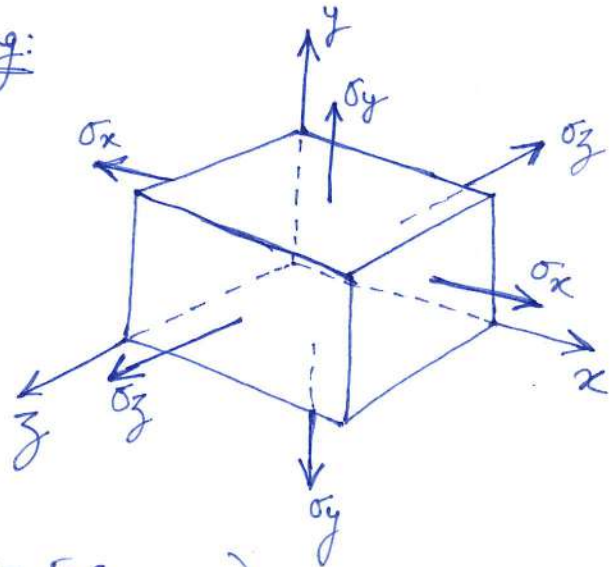
$$\Rightarrow \epsilon_v = \frac{\delta L}{L} + \left(-\mu \frac{\delta L}{L}\right) + \left(-\mu \frac{\delta L}{L}\right)$$

$$\boxed{\epsilon_v = \frac{\delta L}{L} (1 - 2\mu)}$$

here, if $\mu = 0.5$, then $\epsilon_v = 0$ (No change in volume).

(b) In case of three dimensional loading:

Let, an elastic block is subjected to tensile stresses σ_x along x-axis, σ_y along y-axis and σ_z along z-axis.



longitudinal strain along
x-axis = $\frac{\sigma_x}{E}$

$$\left(\sigma = E \cdot \epsilon \right. \\ \left. \Rightarrow \epsilon = \frac{\sigma}{E} \right)$$

longitudinal strain along y-axis
= $\frac{\sigma_y}{E}$

and longitudinal strain along z-axis = $\frac{\sigma_z}{E}$

Therefore, after considering lateral strains also,

$$\text{Total effective strain along x-axis} = \frac{\sigma_x}{E} - \mu \cdot \frac{\sigma_y}{E} - \mu \frac{\sigma_z}{E} \quad \text{--- (I)}$$

$$\text{Total effective strain along y-axis} = \frac{\sigma_y}{E} - \mu \cdot \frac{\sigma_z}{E} - \mu \cdot \frac{\sigma_x}{E} \quad \text{--- (II)}$$

$$\text{and total effective strain along z-axis} = \frac{\sigma_z}{E} - \mu \cdot \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E} \quad \text{--- (III)}$$

by adding (I), (II) and (III),

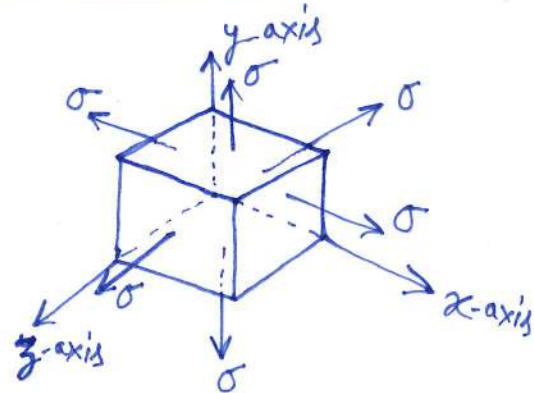
$$\text{Vol. Strain } \epsilon_v = \epsilon_x + \epsilon_y + \epsilon_z$$

$$\Rightarrow \epsilon_v = (1 - 2\mu) \frac{\sigma_x}{E} + (1 - 2\mu) \frac{\sigma_y}{E} + (1 - 2\mu) \frac{\sigma_z}{E}$$

$$\boxed{\epsilon_v = (1 - 2\mu) \left(\frac{\sigma_x + \sigma_y + \sigma_z}{E} \right)} \quad \text{--- (IV)}$$

⑮ Relation between Young's Mod. and Bulk Modulus:-

Let, there is a block under hydrostatic stress system. Means that the stresses along x, y, and z-axes are same (σ).



$$\sigma_x = \sigma_y = \sigma_z = \sigma = \text{volumetric stress.}$$

$$\Rightarrow \text{Total effective strain along x-axis} = \frac{\sigma}{E} - \mu \frac{\sigma}{E} - \mu \frac{\sigma}{E} \\ = \frac{\sigma}{E} (1 - 2\mu)$$

$$\text{Similarly, total effective strain along y-axis} = \frac{\sigma}{E} (1 - 2\mu) \\ \text{and total effective strain along z-axis} = \frac{\sigma}{E} (1 - 2\mu)$$

$$\Rightarrow \text{Vol. Strain } \epsilon_v = \left\{ \frac{\sigma}{E} (1 - 2\mu) \right\} + \left\{ \frac{\sigma}{E} (1 - 2\mu) \right\} + \left\{ \frac{\sigma}{E} (1 - 2\mu) \right\}$$

$$\epsilon_v = \frac{3\sigma}{E} (1 - 2\mu) \quad \text{--- (A)}$$

but, by definition of Bulk Modulus,

$$K = \frac{\text{Vol. Stress}}{\text{Vol. Strain}} = \frac{\sigma}{\epsilon_v} \Rightarrow \epsilon_v = \frac{\sigma}{K} \quad \text{--- (B)}$$

Comparing (A) and (B),

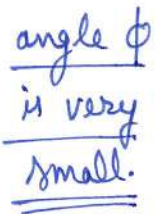
$$\cancel{\frac{\sigma}{K}} \nearrow \frac{3\cancel{\sigma}}{E} (1 - 2\mu)$$

$$\boxed{E = 3K(1 - 2\mu)}$$

⑯ Relation Between Young's Mod. and Mod. of Rigidity:-

Consider a square elastic block ADCB, under the action of shear stresses on the planes AB and DC. The "depth" of the block is unity (i.e. the dimension $\perp$ the plane of paper). Complimentary shear stresses develop on the vertical surfaces AD and CB

The block distorts into a new shape $AD'C'B$. One diagonal elongates ($AC \rightarrow AC'$) and the other diagonal shrinks ($DB \rightarrow D'B$)


$$= \frac{AC' - AC}{AC}$$

$$= \frac{AC' - AE}{AC} \quad (\because AC \approx AE)$$

Now, $\angle CC'E \approx 45^\circ$

$$\Rightarrow EC' = CC' \cos 45^\circ = \frac{CC'}{\sqrt{2}}$$

$$\Rightarrow \text{Longitudinal strain in diagonal AC} = \frac{CC'}{\sqrt{2} \cdot AC} \quad [\text{from ①}]$$

$$= \frac{CC'}{\sqrt{2} \cdot (BC\sqrt{2})} = \frac{CC'}{2 \cdot BC} \quad \text{--- (11)}$$

but $\frac{CC'}{BC} = \text{shear strain } \tan \phi \approx \phi,$

$$\text{and } \phi = \frac{\text{shear stress}}{\text{mod. of rigidity}} = \frac{\tau}{C}$$

$\Rightarrow$ from (II), Longitudinal strain in the diagonal AC

$$= \frac{\tau}{2C} \text{ --- (III)}$$

Now, Considering the half diagram of the above mentioned block.
(BCD)

Force on the surface $DC = T \times DC \times 1 \quad (\rightarrow)$

Force on the surface BC = $\tau \times BC \times 1$ ($\uparrow$)

Force on the surface BD = $\sigma_t \times BD \times 1$ (✓)

Under the conditions of equilibrium,

$$(\sigma_t \times BD \times 1) \cos 45^\circ = (\tau \times BC \times 1)$$

$$\Rightarrow \sigma_t \times (BC\sqrt{2}) \times \cos 45^\circ = \tau \times BC$$

$$\sigma_t \times \sqrt{2} \times \frac{1}{\sqrt{2}} = \tau$$

$$\text{or } \sigma_t = \tau. \text{ [Similarly, } \sigma_c = \tau \text{]}$$

putting this value in equation (III), we get

$$\text{Longitudinal strain in the diagonal AC} = \frac{\sigma_t}{2C} \text{ ————— (IV)}$$

But strain in diagonal AC is also given by —

= strain due to tensile stress in AC — strain due to compressive stress in BD

$$= \frac{\sigma_t}{E} - \left(-\mu \frac{\sigma_t}{E}\right) = \frac{\sigma_t}{E} (1+\mu) \text{ ————— (V)}$$

Comparing (IV) and (V),

$$\frac{\sigma_t}{2C} = \frac{\sigma_t}{E} (1+\mu)$$

$$\Rightarrow \boxed{E = 2C(1+\mu)}$$

(17) Relation Between all the elastic constants E, C, K —

$$E = 2C(1+\mu) \Rightarrow \frac{E}{2C} = 1+\mu \text{ or } \mu = \frac{E}{2C} - 1$$

replacing the value of μ in the relation $E = 3K(1-2\mu)$

$$\Rightarrow E = 3K \left(1 - 2\left(\frac{E}{2C} - 1\right)\right) = 3K \left(1 - \frac{2E - 4C}{2C}\right)$$

$$= 3K \left(\frac{2C - 2E + 4C}{2C}\right) = 3K \left(\frac{6C - 2E}{2C}\right)$$

$$E = 3K \left(\frac{3C - E}{C} \right)$$

$$\Rightarrow EC = 9KC - 3KE$$

$$\Rightarrow (3K + C)E = 9KC$$

$$\Rightarrow \boxed{E = \frac{9KC}{3K + C}}$$

(18) NUMERICAL PROBLEMS

Q. A steel bar is 900 mm long; its two ends are 40 mm and 30 mm in diameter and the length of each rod is 200 mm. The middle portion of the bar is 15 mm in diameter and 500 mm long. If the bar is subjected to an axial tensile load of 15 kN, find the total extension.

$$E = 200 \text{ GPa}$$

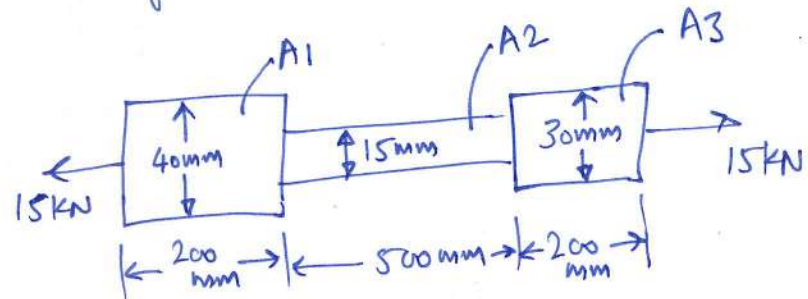
Sol.

$$P = 15 \text{ kN}$$

$$A_1 = \frac{\pi}{4} \times 40^2 = 1256.6 \text{ mm}^2$$

$$A_2 = \frac{\pi}{4} \times 15^2 = 176.7 \text{ mm}^2$$

$$A_3 = \frac{\pi}{4} \times 30^2 = 706.8 \text{ mm}^2$$



$$\delta l_1 = \frac{Pl_1}{A_1 E} = \frac{15 \times 10^3 \times 200}{1256.6 \times 200 \times 10^3} = 0.0119 \text{ mm.}$$

$$\delta l_2 = \frac{Pl_2}{A_2 E} = \frac{15 \times 10^3 \times 500}{176.7 \times 200 \times 10^3} = 0.212 \text{ mm.}$$

$$\delta l_3 = \frac{Pl_3}{A_3 E} = \frac{15 \times 10^3 \times 200}{706.8 \times 200 \times 10^3} = 0.021 \text{ mm}$$

$$\Rightarrow \text{Total elongation} = \delta l_1 + \delta l_2 + \delta l_3$$

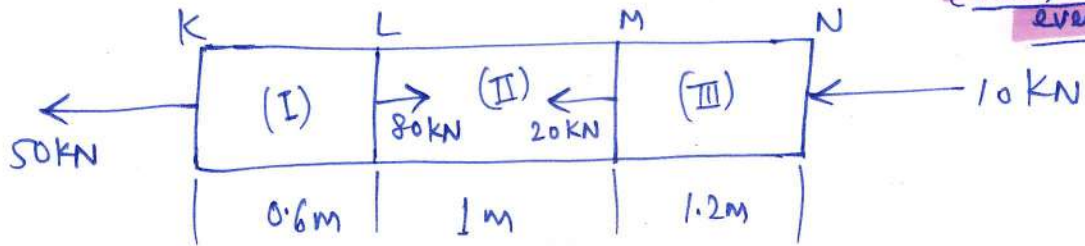
$$= 0.2449 \text{ mm} \quad \underline{\underline{\text{Ans.}}}$$

here GPa has been converted into MPa, because $\text{MPa} = \text{N/mm}^2$

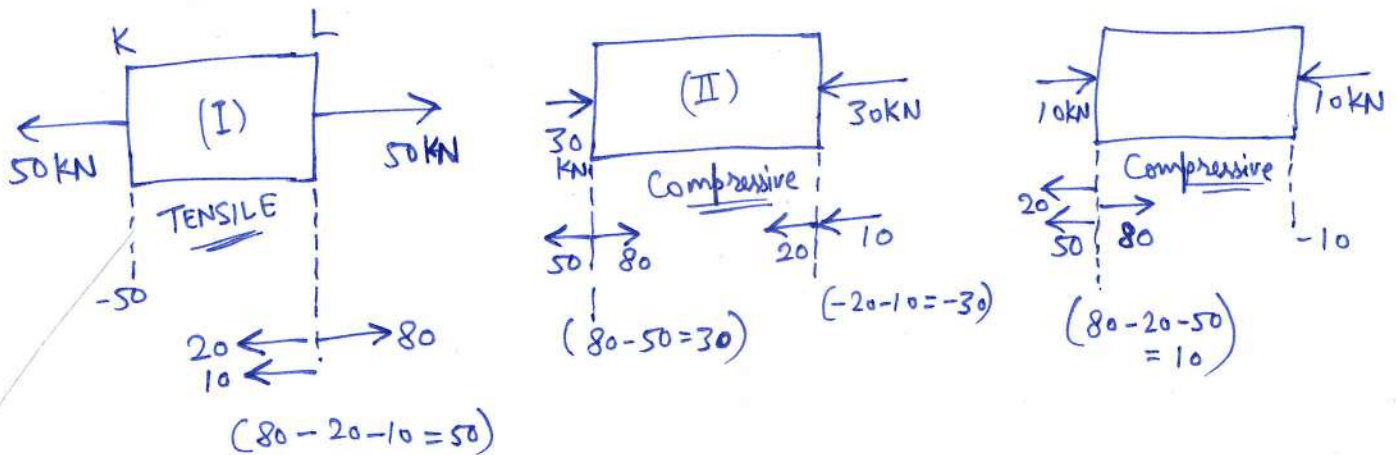
Q. A brass bar having cross-sectional area of 1000 mm^2 is subjected to axial forces shown in the figure. Find the total elongation of the bar. $E_{\text{brass}} = 100 \text{ GPa}$.

(AKTU, 2022-23, even, C.O.)

Solⁿ



Drawing the individual FBDs of the portions KL, LM and MN-



Net elongation of the bar:

$$\delta l = \delta l_I + \delta l_{II} + \delta l_{III}$$

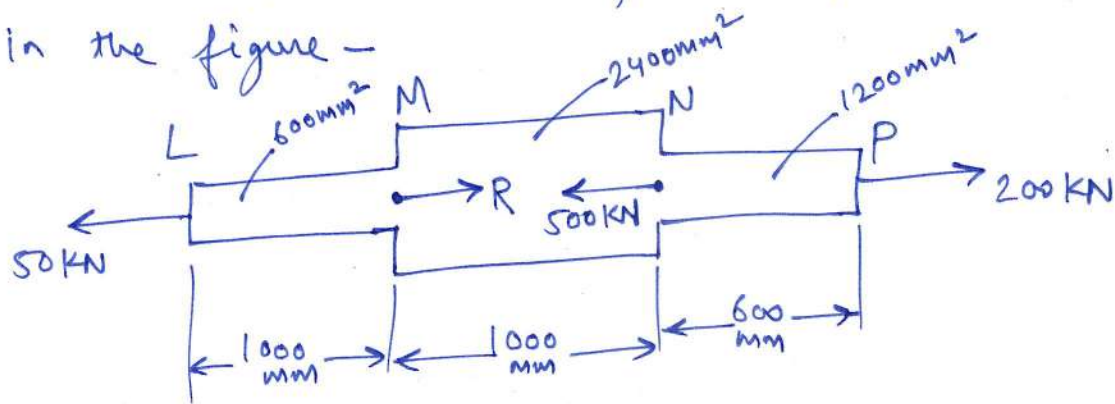
$$= \frac{P_1 l_1}{A_1 E} + \frac{P_2 l_2}{A_2 E} + \frac{P_3 l_3}{A_3 E}$$

$$= \frac{50 \times 10^3 \times 600}{1000 \times 1000 \times 10^3} + \frac{(-30 \times 10^3 \times 1000)}{1000 \times 1000 \times 10^3} + \frac{(-10 \times 10^3 \times 1200)}{1000 \times 1000 \times 10^3}$$

$$= 0.3 - 0.3 - 0.12 = -0.12 \text{ mm}$$

$$= \boxed{0.12 \text{ mm Compression}} \text{ Ans.}$$

Q. A member LMNP is subjected to point loads as shown in the figure -

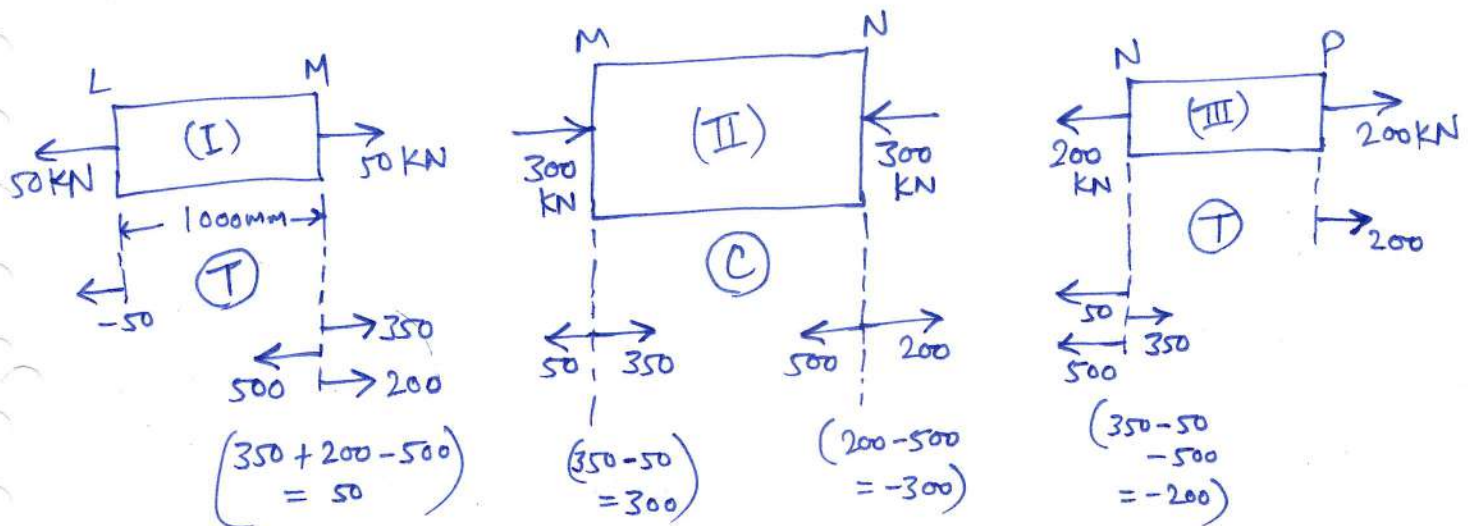


Calculate: (a) Force R necessary for equilibrium, (b) total elongation of the bar, Take $E = 210 \text{ GPa}$.

Solⁿ... Drawing the individual FBDs of the three parts -

$$\text{here, } -50 + R - 500 + 200 = 0$$

$$\Rightarrow \boxed{R = 350 \text{ kN}} \text{ Ans.}$$



$$\text{Net elongation } \delta l = \delta l_I + \delta l_{II} + \delta l_{III}$$

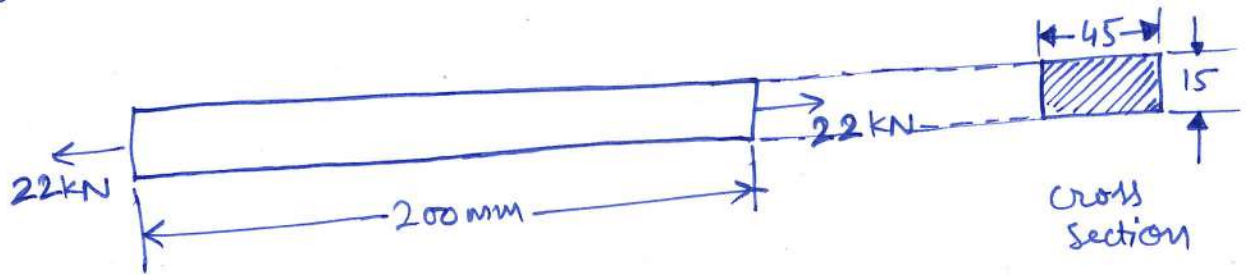
$$\delta l = \frac{50 \times 10^3 \times 1000}{600 \times 210 \times 10^3} + \frac{-300 \times 10^3 \times 1000}{2400 \times 210 \times 10^3} + \frac{200 \times 10^3 \times 600}{1200 \times 210 \times 10^3}$$

$$= 0.397 - 0.595 + 0.476$$

$$= \boxed{0.278 \text{ mm}} \text{ Ans.}$$

- Q. At an axial load of 22 kN, a 45 mm wide by 15 mm thick polyimide bar elongates 3.0 mm while the bar width contracts 0.25 mm. The bar is 200 mm long. At the 22 kN load, the stress in the polymer bar is less than its proportional limit. Determine -
 (a) mod. of elasticity, (b) poisson's Ratio, (c) Change in the bar thickness.
 (AKTU, 2022-23, odd sem)

Sol.ⁿ



$$F = 22 \text{ kN}, L = 200 \text{ mm}, \Delta l = 3 \text{ mm}, \Delta w = 0.25 \text{ mm}, w = 45 \text{ mm}$$

$$\Delta l = \frac{FL}{AE} \Rightarrow E = \frac{FL}{A \cdot \Delta l} = \frac{22 \times 10^3 \times 200}{(45 \times 15) \times 3}$$

$$\Rightarrow E = 2172.83 \text{ N/mm}^2 \quad \text{Ans.}$$

$$\text{Lateral strain} = \frac{\Delta w}{w} = \frac{0.25}{45} = 0.00555$$

$$\text{Longitudinal strain} = \frac{\Delta l}{L} = \frac{3.0}{200} = 0.015$$

$$\Rightarrow \mu = \frac{\text{Lat. Strain}}{\text{Long. Strain}} = \frac{0.00555}{0.015} = 0.37 \quad \text{Ans.}$$

$$\text{Also, lateral strain} = \frac{\Delta t}{t}$$

$$0.00555 = \frac{\Delta t}{15}$$

$$\Rightarrow \Delta t = 15 \times 0.00555 = 0.083 \text{ mm} \quad \text{Ans.}$$

Q. Consider the following data —
diameter of a bar = 30 mm

Tensile load = 54 kN

gauge length = 300 mm

Extension of the bar = 0.112 mm

Change in dia = 0.00366

Calculate Poisson's ratio, mod. of elasticity, mod. of rigidity and Bulk modulus.
(AKTU, 2022-23, even sem)

Solⁿ

$$E = \frac{FL}{A \cdot \Delta L} = \frac{54 \times 10^3 \times 300}{\left(\frac{\pi}{4} \times 30^2\right) \times 0.112} = \boxed{204627.783 \text{ N/mm}^2}$$

$$\mu = \frac{\text{lat. strain}}{\text{long. strain}} = \frac{\Delta d/d}{\Delta L/L} = \frac{\Delta d}{d} \times \frac{L}{\Delta L} = \frac{0.00366}{30} \times \frac{300}{0.112}$$
$$= \boxed{0.33 \text{ Ans.}}$$

$$E = 3K(1 - 2\mu) \Rightarrow K = \frac{E}{3(1 - 2\mu)} = \frac{204627.783}{3(1 - 2 \times 0.33)}$$
$$= \boxed{200615.47 \text{ N/mm}^2}$$

$$E = 2C(1 + \mu) \Rightarrow C = \frac{E}{2(1 + \mu)} = \frac{204627.783}{2(1 + 0.33)}$$
$$= \boxed{76927.74 \text{ N/mm}^2}$$

Q. A metallic wire ($E = 20 \times 10^{10} \frac{\text{N}}{\text{m}^2}$, $\mu = 0.26$) of length 3 m and diameter 0.1 cm is stretched by a load of 10 kg. Calculate the decrease in diameter of the wire.
(AKTU, 2023-24 odd)

Solⁿ

$$E = 20 \times 10^{10} \text{ N/m}^2, \mu = 0.26, l = 3 \text{ m}, d = 0.1 \times 10^{-2} \text{ m}$$

$$F = 10 \text{ kg} \times 9.81 = 98.1 \text{ N}$$

$$\Delta l = \frac{FL}{AE} = \frac{98.1 \times 3}{\frac{\pi}{4} (0.1 \times 10^{-2})^2 \times 20 \times 10^9} = 0.0018 \text{ m}$$

$$\mu = \frac{\Delta d/d}{\Delta l/L} \Rightarrow \mu = \frac{\Delta d}{d} \times \frac{L}{\Delta l}$$

$$\text{or } \Delta d = \frac{\mu \cdot d \cdot \Delta l}{L} = \frac{0.26 \times 0.001 \times 0.0018}{3}$$

$$= 0.000000156 = \boxed{1.56 \times 10^{-7} \text{ m}} \text{ Ans.}$$

Q. A steel bar of square cross-section 35 mm x 35 mm, 500 mm long stretches 0.2 mm under a pull of 100 kN. The same bar in a single shear test under a force of 122.5 kN shows a distortion of original right angled corners by 0.00125 radians. Determine the values of 4 elastic constants.

Solⁿ. $E = \frac{FL}{A \cdot \Delta L} = \frac{100 \times 10^3 \times 500}{(35 \times 35) \times 0.2} = \boxed{204081.63 \text{ N/mm}^2}$

$$\text{mod. of rigidity } C = \frac{\text{shear stress}}{\text{shear strain}} = \frac{\text{shearing force/area}}{\text{shear strain}}$$

here, shearing force = 122.5 kN, and

shearing strain = angular distortion of right angles = 0.00125 rad.

$$= C = \frac{122.5 \times 10^3 / (35 \times 35)}{0.00125} = \boxed{80000 \text{ N/mm}^2}$$

$$\text{Now, } E = 2C(1+\mu) \Rightarrow \mu = \frac{E}{2C} - 1 = \frac{204081.63}{2 \times 80000} - 1$$

$$\boxed{\mu = 0.27}$$

$$\text{Also, } K = \frac{E}{3(1-2\mu)} = \frac{204081.63}{3(1-2 \times 0.27)}$$

$$K = 147885.24 \text{ N/mm}^2 \quad \underline{\text{Ans.}}$$

Q. A C.I flat, 300 mm long and 30 mm x 50 mm uniform section, is acted upon by the following forces over the respective cross-sections $\rightarrow$ 25 kN in the direction of length (tensile), 350 kN in the direction of width (compressive) and 200 kN in the direction of thickness (tensile). Determine the change in volume of the flat. Take $E = 140 \text{ GPa}$, $\mu = 0.25$

Solⁿ. $\sigma_x = \frac{-350 \times 10^3}{(30 \times 300)} = -38.889 \text{ N/mm}^2$

$$\sigma_y = \frac{25 \times 10^3}{(30 \times 50)} = 16.667 \text{ N/mm}^2$$

$$\sigma_z = \frac{200 \times 10^3}{(50 \times 300)} = 13.333 \text{ N/mm}^2$$

Net strain along x-axis $\rightarrow$

$$\epsilon_x = \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E} - \mu \frac{\sigma_z}{E}$$

$$= \frac{1}{140 \times 10^3} [-38.889 - 0.25 \times 16.667 - 0.25 \times 13.333]$$

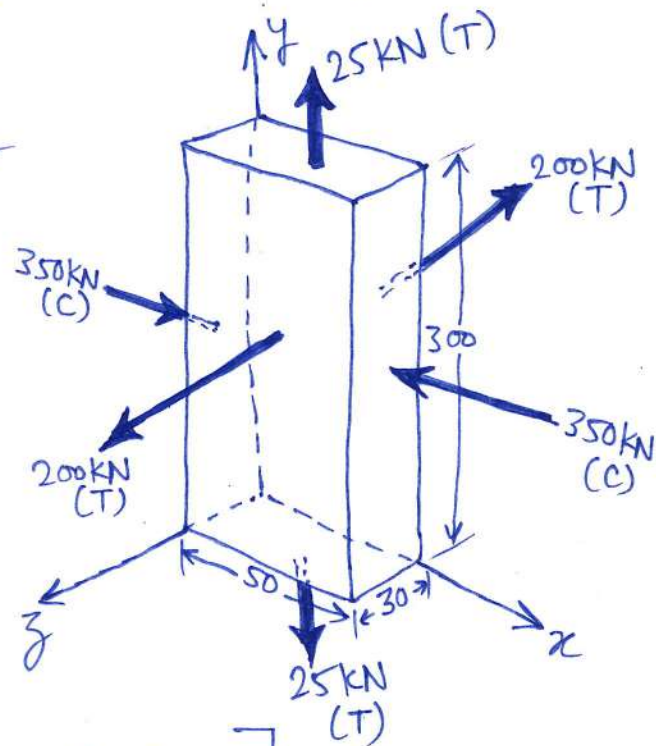
$$= -0.000331$$

Similarly, Net strain along y-axis $\rightarrow$

$$\epsilon_y = \frac{\sigma_y}{E} - \mu \frac{\sigma_z}{E} - \mu \frac{\sigma_x}{E}$$

$$= \frac{1}{140 \times 10^3} [16.667 - 0.25 \times 13.333 - 0.25 \times (-38.889)]$$

$$= 0.000165$$



$$\begin{aligned}
 \epsilon_z &= \frac{\sigma_z}{E} - \mu \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E} \\
 &= \frac{1}{140 \times 10^3} [13.333 - 0.25 \times (-38.889) - 0.25 \times 16.667] \\
 &= 0.000135
 \end{aligned}$$

$$\Rightarrow \text{Volumetric Strain } \epsilon_v = \epsilon_x + \epsilon_y + \epsilon_z$$

$$\epsilon_v = -0.000331 + 0.000165 + 0.000135$$

$$= -0.000031 \quad (\text{indicates overall decrease in volume})$$

$$\frac{\delta V}{V} = \text{Vol. Strain}$$

$$\Rightarrow \text{Change in Volume} = -0.000031 \times (300 \times 50 \times 30)$$

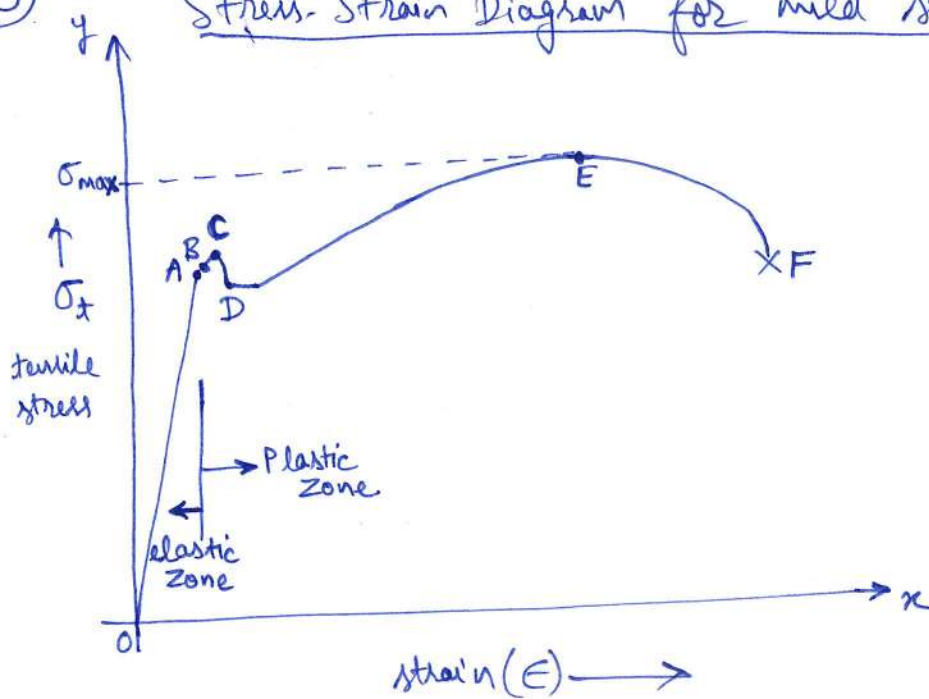
$$= \boxed{-13.95 \text{ mm}^3}$$

volume
of the flat

i.e. Overall decrease in volume of the given block = 13.95 mm³ Ans.

(19)

Stress-Strain Diagram for mild steel, and other Ductile materials.



S.S. Curve for mild Steel

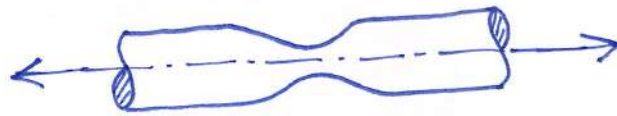
When tensile test is run on UTM on specially designed test specimen, stress and strain data is obtained, which is plotted on xy -coordinate system. The salient points on the graph are given below —

- 1) Initially stress $\propto$ strain, hence straight line is obtained upto a point A, known as "Limit of Proportionality".
- 2) Then a point B is obtained very close to point A. The material loses its property of elasticity after this point. It is called "elastic limit".
- 3) The curvature increases further upto point C, from which it suddenly drops to a point D. These points C and D are called upper and lower yield points respectively. Yield point is a point on SS curve from which larger, clear and permanent deformations take place. The length increases with a much faster rate after yield point.

- 4) After yield points, the graph rises again but with slower rate of increase in stress. Finally a maximum stress-point is obtained (point E) known as "Ultimate tensile Strength" (σ_{max}).

The portion DE is called 'Strain Hardening'

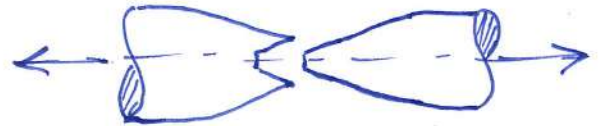
- 5) After the point E, the graph drops down again, and "Neck formation" or "Necking" takes place.



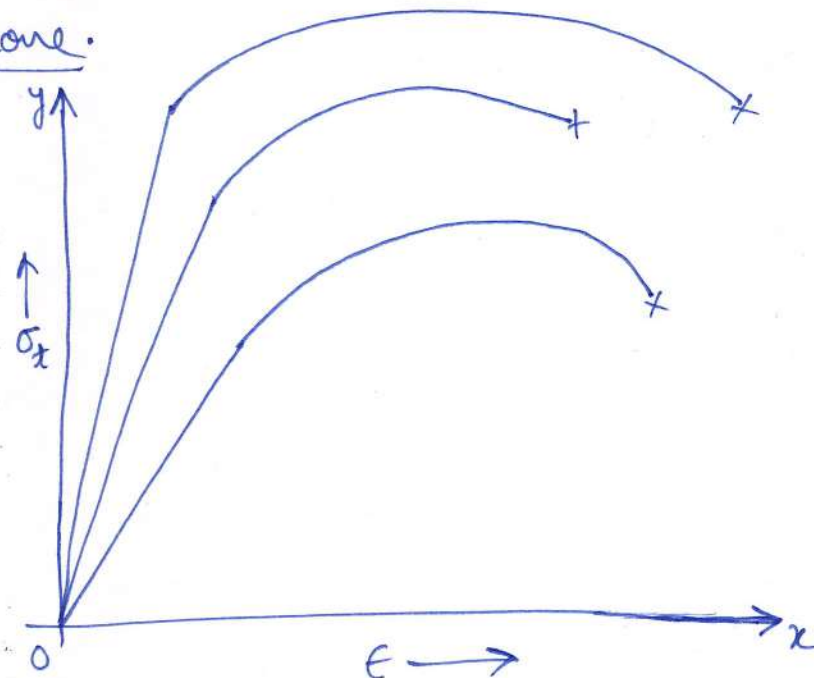
the cross-section area drastically drops at a particular location.

- 6) In the end, the test piece breaks in two parts. A cup and cone type of fracture is observed.

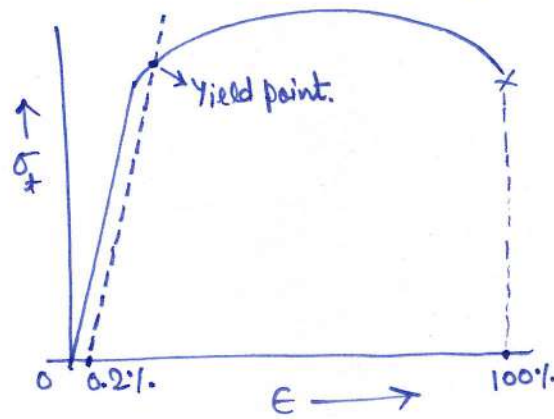
- 7) The part of the curve before point B is called elastic zone, and the part after point B is called plastic zone or permanent deformation zone.



- 8) For the ductile materials other than mild steel, generally yield point is not identifiable easily, and a smooth curve is obtained as shown —



For locating yield point, 0.2% of total strain is marked on the x-axis near origin, and a line is drawn parallel to the initial straight line part of SS curve. This line intersects the SS curve at the yield point—



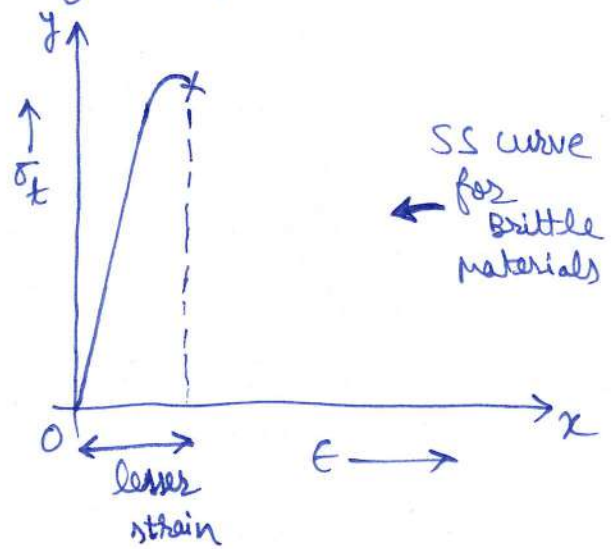
SS curve for
ductile materials
other than
mild steel.

② Stress-Strain Diagram for Brittle materials:—

Brittle materials show very small permanent deformations, and when the tensile load is increased gradually, they break suddenly without showing large deformations.

Initially, stress is proportional to strain for brittle materials also. But after that part, the curve does not expand much along x-axis.

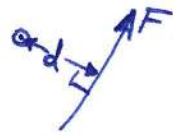
A sudden, flat fracture is observed in brittle materials without any necking part.



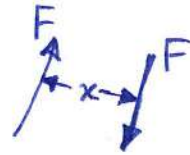
SS curve
for
brittle
materials

② Unit Summary and Important Formulas

- 1) Moment of a force $M = F \times d$
 $d = \perp$ distance of force



- 2) Couple is $C = F \cdot x$
 x is distance between two equal and opposite forces.



- 3) Resultant of two forces having angle θ between them -
 $R = \sqrt{F_1^2 + F_2^2 + 2 F_1 F_2 \cos \theta}$

angle of resultant (from F_2) $\beta = \tan^{-1} \left(\frac{F_1 \sin \theta}{F_2 + F_1 \cos \theta} \right)$

- 4) Resultant of two forces having angle 90° between them -

$$R = \sqrt{F_1^2 + F_2^2}$$

and angle of resultant (from F_2) $\beta = \tan^{-1} \left(\frac{F_1}{F_2} \right)$

- 5) Resultant of several forces in a coplanar force system -

$$R = \sqrt{(\sum F_H)^2 + (\sum F_V)^2}$$

$$\beta = \tan^{-1} \left(\frac{\sum F_V}{\sum F_H} \right)$$

F_H = Sum of all Horizontal components of forces,

F_V = Sum of all vertical component of forces.

- 6) Conditions of static equilibrium in a coplanar forces system:-

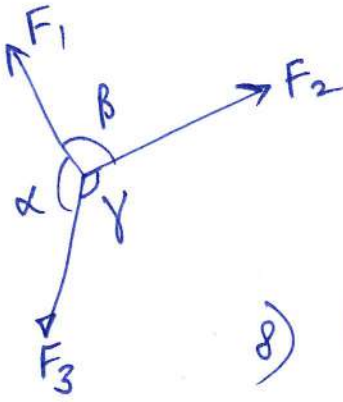
- $\sum F_x = 0$
- $\sum F_y = 0$
- $\sum M = 0$

(i.e. $\sum F_H = 0$)

(i.e. $\sum F_V = 0$)

(moment of all forces about any point is equal to zero).

7) Lami's theorem :- Applicable for 3 coplaner, concurrent forces in static equilibrium $\rightarrow$



$$\frac{F_1}{\sin \gamma} = \frac{F_2}{\sin \alpha} = \frac{F_3}{\sin \beta} = \text{constant}$$

8) Condition of Concurrency :- If three forces are in static equilibrium,

then the lines of action of these forces must pass through a single point (for coplaner force system)

9) Varignon's Theorem :- For coplaner force system,

$$F_1 \cdot d_1 + F_2 \cdot d_2 + F_3 \cdot d_3 + \dots + F_n \cdot d_n = R \cdot d$$

$$\sum_{i=1}^n F_i \cdot d_i = R \cdot d$$

i.e. Sum of individual moments = moment of the resultant

Beams

1) types of load - point load, UDL, UVL, point moment

2) types of supports - simple support, roller support, hinge support, Fixed support

3) types of Beams

- $\rightarrow$ Statically determinate $\rightarrow$ Simply supported, Cantilever
(No. of unknown reactions are 3 or less)
Few cases of Over-Hanging.
- $\rightarrow$ Statically Indeterminate $\rightarrow$ Propped Cantilever, Continuous Beams, Both side fixed.
(No of Unknown reactions are more than 3)

Stress and Strain

1) Stress $\sigma = \frac{F}{A}$ (N/m^2 or 'Pa')

2) Strain $\epsilon = \frac{\text{Change in dimension}}{\text{initial dimension}}$ (Unitless)

$$1 \text{ KPa} = 10^3 \text{ Pa}, \quad 1 \text{ MPa} = 10^6 \text{ Pa}, \quad 1 \text{ GPa} = 10^9 \text{ Pa}.$$

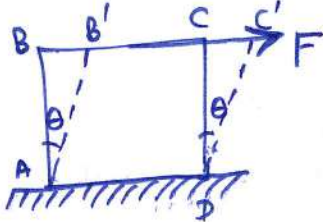
$$1 \text{ N/mm}^2 \equiv 1 \text{ MPa}, \quad 1 \text{ kN/mm}^2 \equiv 1 \text{ GPa}$$

3) Normal Stress = $\frac{\text{Force } \perp \text{ to the cross section}}{\text{Cross-sectional area}}$
(Longitudinal stress)

Normal strain = $\frac{\text{Change in the dimension along the applied load}}{\text{Original dimension}}$
(Longitudinal strain) $\left(\frac{\Delta L}{L}\right)$

4) Shear stress (τ) = $\frac{\text{Tangential force}}{\text{Cross-sectional area}}$ OR $\frac{\text{Force } \parallel \text{ to the surface}}{\text{Surface area}}$

Shear strain (θ) = Angular distortion due to tangential forces



$$\begin{aligned} \text{shear strain} &= \tan \theta \approx \theta \\ &= \frac{BB'}{AB} \quad \text{OR} \quad \frac{CC'}{CD} \end{aligned}$$

5) Volumetric Stress = Hydrostatic 3 dimensional stress
 $= p$

$$\text{volumetric strain} = \frac{\Delta V}{V} = \frac{\text{Change in Volume}}{\text{Original Volume}}$$

6) Hooker's Law : Within the elastic limits,

$$\sigma \propto \epsilon \quad \text{OR} \quad \boxed{\sigma = E \cdot \epsilon}$$

E = Modulus of elasticity.

7) Young's Modulus (E) or (Y) = $\frac{\text{Longitudinal stress}}{\text{Longitudinal strain}}$

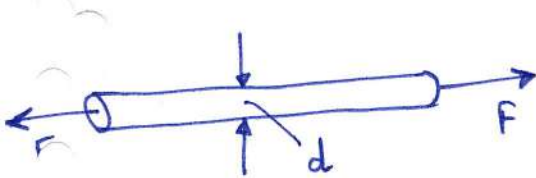
$$E = \frac{F/A}{\Delta L/L} = \frac{FL}{A \cdot \Delta L}$$

or $\Delta L = \frac{FL}{AE}$

8) Mod. of Rigidity 'C' or 'G' = $\frac{\text{Shear stress}}{\text{Shear strain}} = \frac{\tau}{\theta}$

9) Bulk modulus (K) = $\frac{\text{Volumetric Stress}}{\text{Volumetric strain}} = \frac{p}{(\Delta V/V)}$

10) Lateral strain = $\frac{\text{Change in the dimension } \perp \text{ to the applied load}}{\text{initial dimension}}$



$$= \frac{\Delta d}{d} \text{ or } \frac{\Delta w}{w} \text{ or } \frac{\Delta t}{t}$$

depends upon the shape of cross-section.

11) Poisson's Ratio (μ) = $\frac{\text{lateral strain}}{\text{longitudinal strain}}$

range: 0 to 0.5

12) Relation between volumetric strain and longitudinal strain

$$\epsilon_v = \epsilon_x + \epsilon_y + \epsilon_z$$

13) Volumetric strain due to unidirectional loading -

$$\epsilon_v = \frac{\delta L}{L} (1 - 2\mu)$$

14) Net effective strains, in case of 3 dimensional loading -

$$\epsilon_x = \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E} - \mu \frac{\sigma_z}{E}$$

$$\epsilon_y = \frac{\sigma_y}{E} - \mu \frac{\sigma_z}{E} - \mu \frac{\sigma_x}{E}$$

$$\epsilon_z = \frac{\sigma_z}{E} - \mu \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E}$$

and

$$\epsilon_v = (1 - 2\mu) \left(\frac{\sigma_x + \sigma_y + \sigma_z}{E} \right)$$

15) Relation between elastic constants —

$$E = 3K(1 - 2\mu)$$

$$E = 2C(1 + \mu)$$

$$E = \frac{9KC}{3K + C}$$

i) Principle of superposition :— Net elongation of a bar is equal to the sum of elongations of individual segments of the bar

$$\Delta L_{\text{Final}} = \Delta L_1 + \Delta L_2 + \Delta L_3 + \dots$$

values of ΔL_1 , ΔL_2 , ΔL_3 , etc. may be positive or negative.

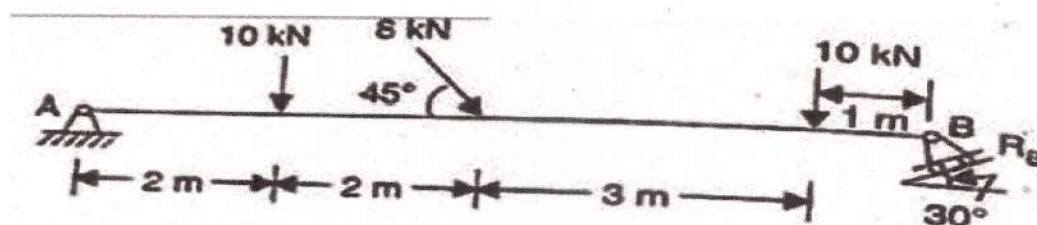
(22)

Important Short-answer type Questions

- 1) What are different types of forces?
- 2) Explain principle of transmissibility.
- 3) What is a Free-body diagram?
- 4) State 'Varignon's theorem'.
- 5) What are the conditions of static equilibrium for a coplanar force system?
- 6) What are statically determinate and statically indeterminate beams?
- 7) What are different force-systems?
- 8) What is shear strain?
- 9) Explain Hooke's Law.
- 10) What is the difference between longitudinal strain and lateral strain?
- 11) Discuss and explain lateral strain.
- 12) What is 'Poisson's Ratio'? Explain.
- 13) Write the relations between all the four ~~to~~ elastic constants.
- 14) What are different kind of supports on a beam?
- 15) Define 'Bulk Modulus' and 'Modulus of Rigidity'.

23**Important Long Answer Type Questions**

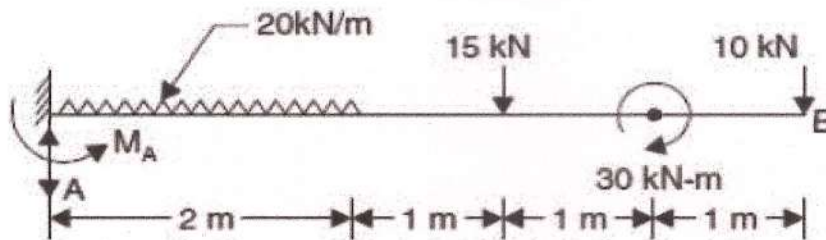
1. Establish the relationship between Modulus of Elasticity and Modulus of Rigidity.
(AKTU End Semester Exam, 2023-24 Even)
2. Derive a relationship between young's Modulus and Bulk Modulus.
3. Find the resultant of forces 2, $\sqrt{3}$, 5, $\sqrt{3}$ and 2 N that act at an angular point of a regular hexagon towards the other angular points taken in order.
(AKTU End Semester Exam, 2023-24 Even)
4. A vertical circular copper bar 20 mm diameter and 3 m long carries a tensile load of 200 kN. Calculate the elongation, decrease in diameter and the volumetric strain. Take $E = 100$ GPa, Poisson's Ratio = 0.25
(AKTU End Semester Exam, 2023-24 Even)
5. A metallic wire ($Y = 20 \times 10^{10}$ N/m² and $\mu = 0.26$) of length 3 m and diameter 0.1 cm is stretched by a load of 10 kg. Calculate the decrease in diameter of the wire.
(AKTU End Semester Exam, 2023-24 Odd)
6. A beam 8 m. long is hinged at A and supported on roller over a smooth surface inclined at an angle 30° to the horizontal at B. The beam is loaded as shown in fig. Determine the support reactions.



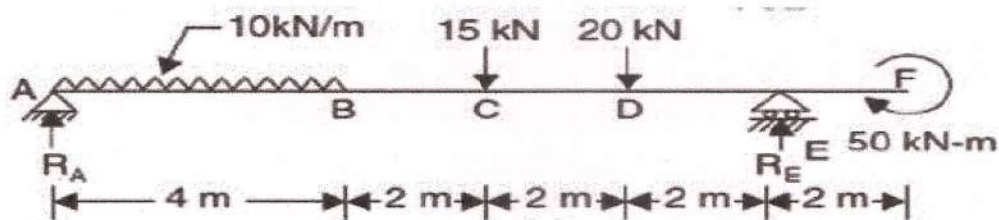
(AKTU End Semester Exam, 2023-24 Odd)

7. What is the shape of cross-section obtained after yielding in cases of brittle material? Explain in brief.
(AKTU End Semester Exam, 2023-24 Odd)
8. The forces 20 N, 30 N, 40 N, 50 N and 60 N are acting at one of the angular points of a regular hexagon, towards the other five angular points, taken in order. Find the magnitude and direction of the resultant force.
(AKTU End Semester Exam, 2022-23 Even)
9. Define Poisson's ratio and bulk modulus. The following data relate to a bar subjected to a tensile test:
 Diameter of the bar = 30 mm
 Tensile load = 54 kN
 Gauge length = 300 mm
 Extension of the bar = 0.112 mm
 Change in diameter = 0.00366 mm
 Calculate Poisson's ratio, modulus of elasticity, modulus of rigidity and bulk modulus.

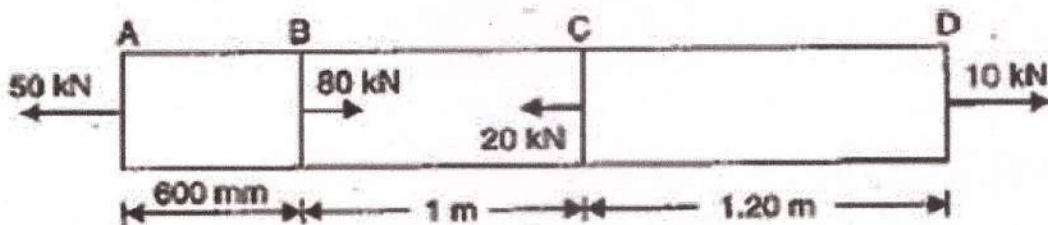
10. Find the support reactions of the cantilever beam loaded as shown in fig.



11. Find the support reactions for the beam loaded as shown in Fig.



12. A brass bar having cross section area of 1000 mm^2 , is subjected to axial force as shown in fig. find the total elongation of the bar take $E = 1.05 \times 10^5 \text{ N/mm}^2$.

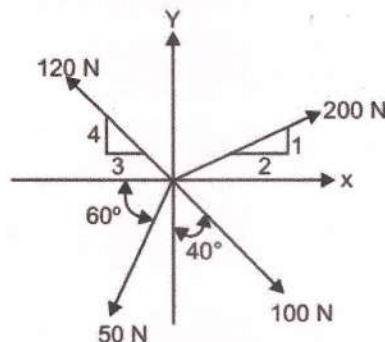


(AKTU End Semester Exam, 2022-23 Even, Carry-Over)

13. Draw and explain the stress strain diagram for the ductile material, and brittle material and for mild steel.

(AKTU End Semester Exam, 2022-23 Odd)

14. A system of four forces acting on a body is as shown in figure. Determine the resultant.



(AKTU End Semester Exam, 2022-23 Odd)

15. At an axial load of 22 kN, a 45-mm-wide by 15-mm thick polyimide polymer bar elongates 3.0 mm while the bar width contracts 0.25 mm. The bar is 200 mm long. At the 22-kN load, the stress in the polymer bar is less than its proportional limit. Determine-

1. The modulus of elasticity.
2. Poisson's ratio.
3. The change in the bar thickness

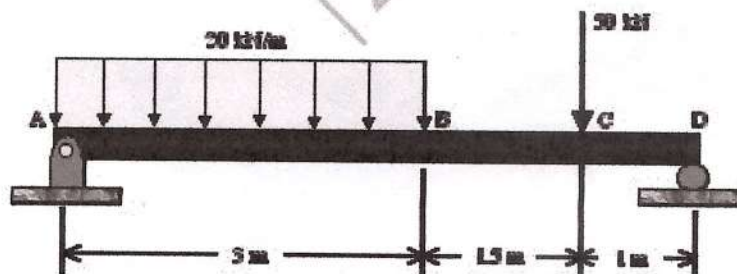
(AKTU End Semester Exam, 2022-23 Odd)

16. A composite bar of uniform cross-section area 200 mm² consists of an aluminum section rigidly fastened between a bronze section and a steel section as shown in figure. Axial loads are applied at the positions indicated. Determine: the Axial load P (kN) to make it in equilibrium, Stress in each section and the total change in length of the bar. Given: $E_{\text{Bronze}} = 100\text{GPa}$, $E_{\text{Al}} = 70\text{GPa}$, $E_{\text{Steel}} = 200\text{GPa}$.



(AKTU End Semester Exam, 2021-22 Even)

17. Determine the support reactions:



18. Develop the relationship between E (Young's modulus), C (Shear modulus), K (Bulk modulus) and μ (Poisson ratio).

(AKTU End Semester Exam, 2021-22 Odd)

19. State and Proof Varignon's Theorem.

20. Explain different types of force systems with suitable examples.

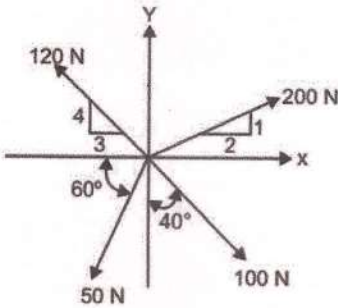
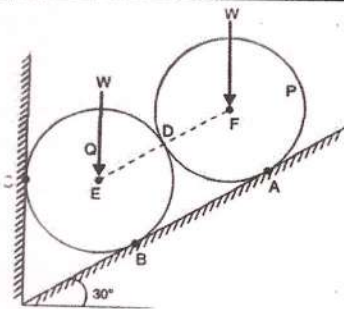
ABES ENGINEERING COLLEGE, GHAZIABAD

Department of Mechanical Engineering

Fundamentals of Mechanical Engineering (BME 101/201)

24

Question Bank Unit-I (Introduction to Mechanics)

Q.NO.	Short Answer type questions
1	Explain principle of transmissibility of force system.
2	Explain Varignon's theorem.
3	What are various types of loads, beams and supports?
4	What is a statically determinate beam? Give some examples too.
5	Define Engineering stress and True stress.
6	How shearing stress is different from normal stress?
7	State and explain Hooke's law and poisson's ratio.
8	Discuss about superposition theorem.
9	What are elastic constants? Give their name.
10	Write formulae for relationship among elastic constants
Q.NO.	Short Answer type questions
1	A system of four forces acting on a body is as shown in figure. Determine the magnitude resultant and its inclination with positive X-axis. 
2	Two identical rollers P and Q, each of weight 100 N, are supported by an inclined plane and a vertical wall as shown in figure. Draw FBD of both sphere assuming all surfaces to be smooth. Also, find the reactions at points A, B, C and D. 
3	Two forces equal to P and 2 P respectively act on a particle. When the first force is increased by 120 N and the second force is doubled, the direction of the resultant remains the same. Determine the value of force P.
4	Two smooth spheres P and Q each of radius 25 cm and weighing 500 N, rest in a horizontal channel having vertical walls. if the distance between the walls is 90 cm, make calculations for the reactions on all the points of contact.

5	Draw and explain stress and strain curve for ductile material. Also, explain the difference between true stress and engineering stress.
6	Illustrate the various important points on stress strain diagram for mild steel. Also, explain the significance of different regions on the same.
7	A brass bar having cross sectional area of 1000 mm^2 is subjected to axial forces as shown in the figure. Find the total elongation of the bar and stress in individual components. Take Modulus of elasticity of Brass = 100 GPa
8	A member ABCD is subjected to point loads P_1, P_2, P_3 and P_4 as shown in figure given below: Calculate the force P_3 necessary for equilibrium if $P_1 = 120 \text{ kN}$, $P_2 = 220 \text{ kN}$ and $P_4 = 160 \text{ kN}$. Determine change in length of the member. Take modulus of elasticity 200 GN/m^2
9	A beam has been loaded and supported as shown in figure given below. Determine the reactions at the support points A and B.
10	Derive the relationship among E (Young's modulus), K (Bulk modulus) and μ (Poisson's ratio).
11	A square bar $50 \text{ mm} \times 50 \text{ mm}$ is subjected to a compressive load of 500 kN . The contraction over 200 mm length is 0.5 mm and increase in thickness is 0.04 mm . Calculate the value of the four elastic constants of the material.
12	At an axial load of 22 kN , a 45-mm -wide by 15-mm thick polyimide polymer bar elongates 3.0 mm while the bar width contracts 0.25 mm . The bar is 200 mm long. At

	the 22-kN load, the stress in the polymer bar is less than its proportional limit. Determine, 1. The modulus of elasticity. 2. Poisson's ratio. 3. The change in the bar thickness
13	A steel bar of square cross-section 35 mm x 35 mm, 500 mm long stretches 0.2 mm under a pull of 100 kN. The same bar in single shear test under a force of 122.5 kN shows a distortion of original right-angle corners by 0.00125 radian. Determine the values of the four elastic constants of the material.